



US 20250275600A1

(19) **United States**

(12) **Patent Application Publication**
POMERING

(10) **Pub. No.: US 2025/0275600 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **CONNECTOR AND APPARATUS**

(52) **U.S. Cl.**

CPC *A42B 3/064* (2013.01)

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(57)

ABSTRACT

(21) Appl. No.: **18/706,427**

(22) PCT Filed: **Oct. 28, 2022**

(86) PCT No.: **PCT/EP2022/080228**

§ 371 (c)(1),

(2) Date: **May 1, 2024**

A connector (20) for connecting first and second parts of an apparatus, the first and second parts being configured to move relative to each other at least in a first plane, the connector comprising: • a first attachment part (21) for connecting to the first part of the apparatus; • a second attachment part (22) for connecting to the second part of the apparatus; • a resilient part (23) extending between the first and second attachment parts and configured to deform to allow relative movement between the first and second attachment parts, wherein: • the first attachment part and the second attachment part are displaced relative to each other in a first direction, and the resilient part is narrower, in a second direction perpendicular to the first direction, than the first attachment part and the second attachment part.

(30) **Foreign Application Priority Data**

Nov. 1, 2021 (GB) 2115653.4

Nov. 1, 2021 (GB) 2115655.9

Publication Classification

(51) **Int. Cl.**

A42B 3/06

(2006.01)

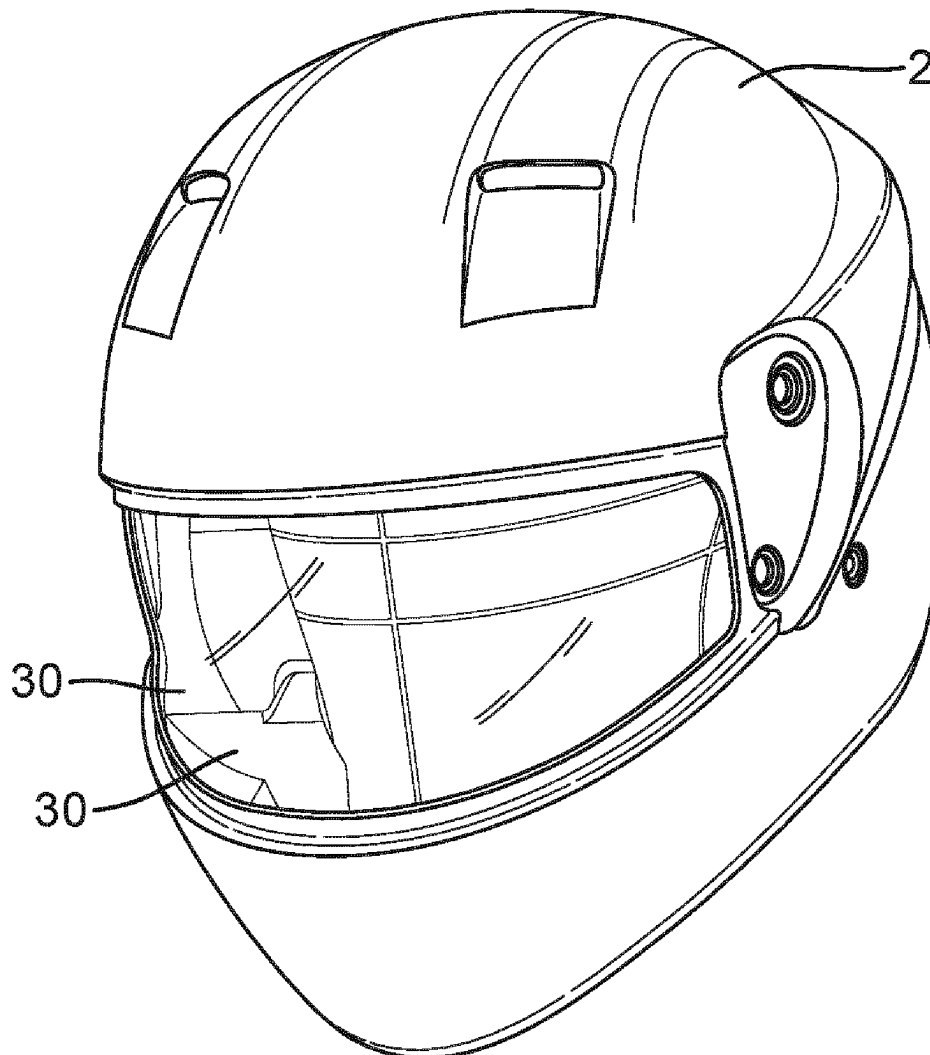


Fig. 1

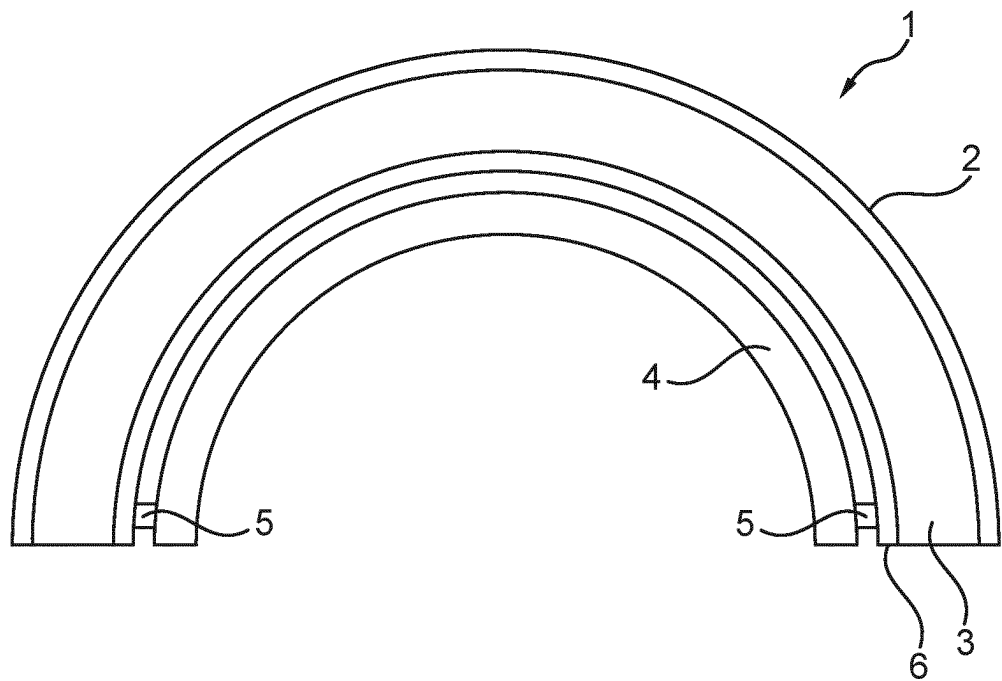


Fig. 2

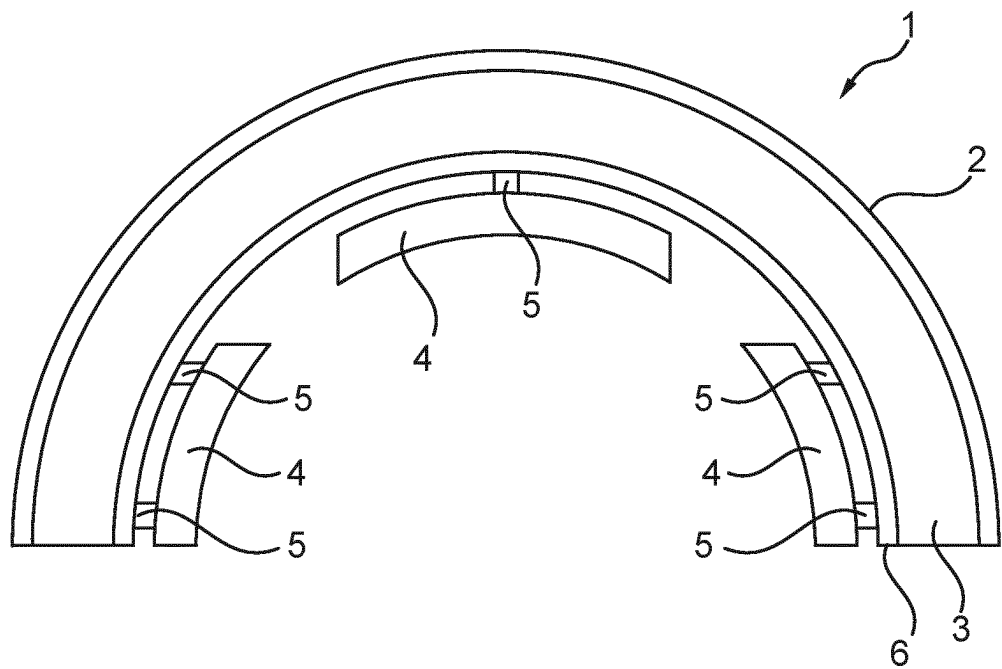


Fig. 3

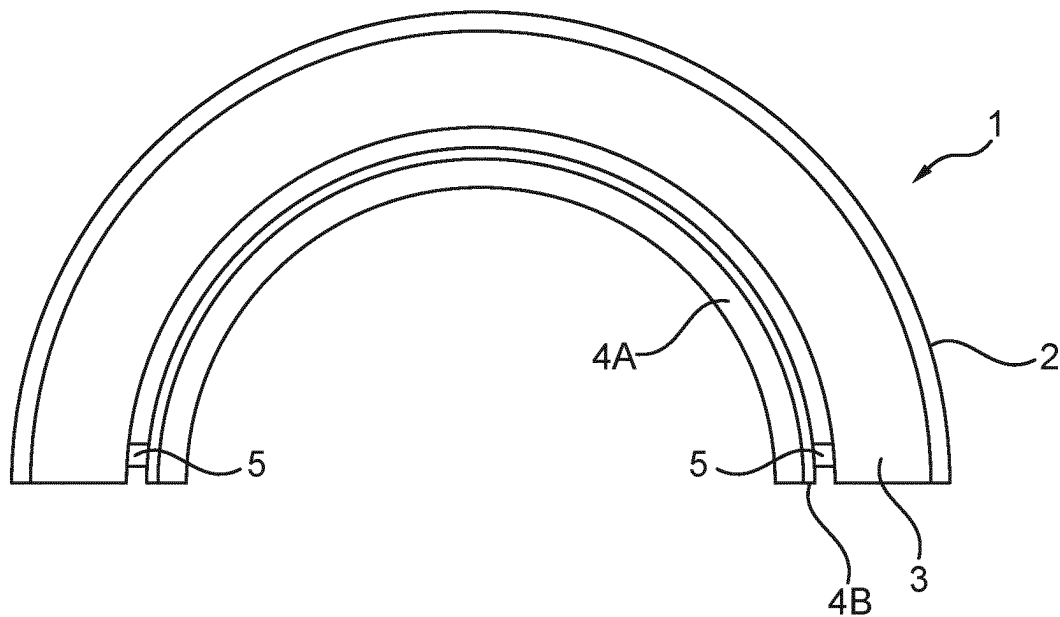


Fig. 4

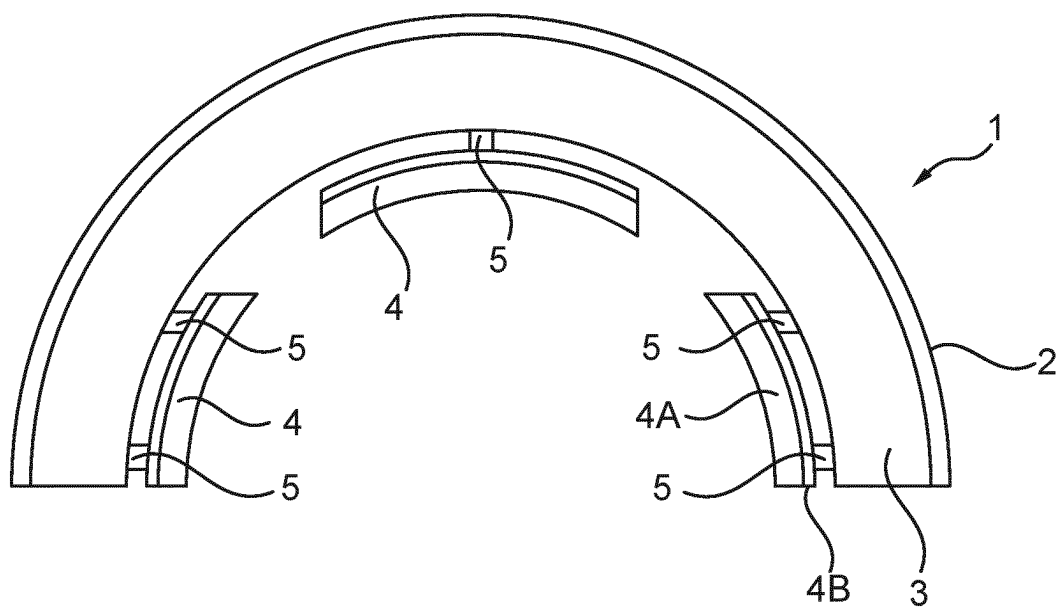


Fig. 5

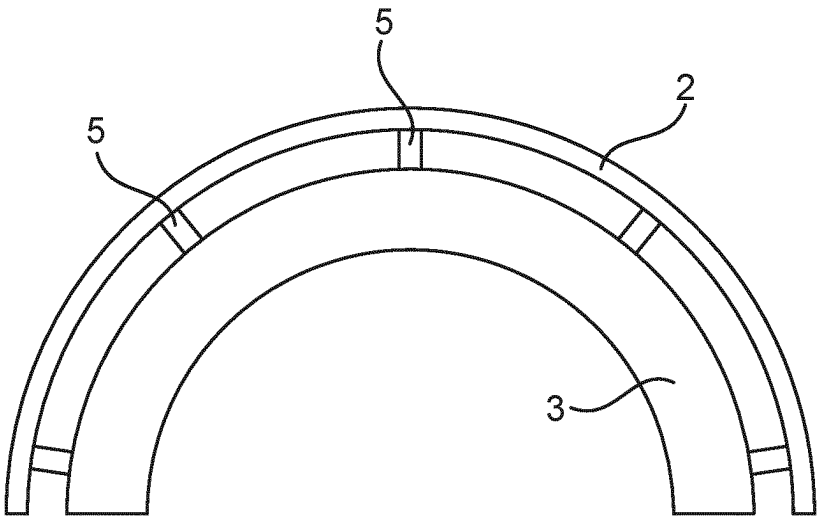


Fig. 6

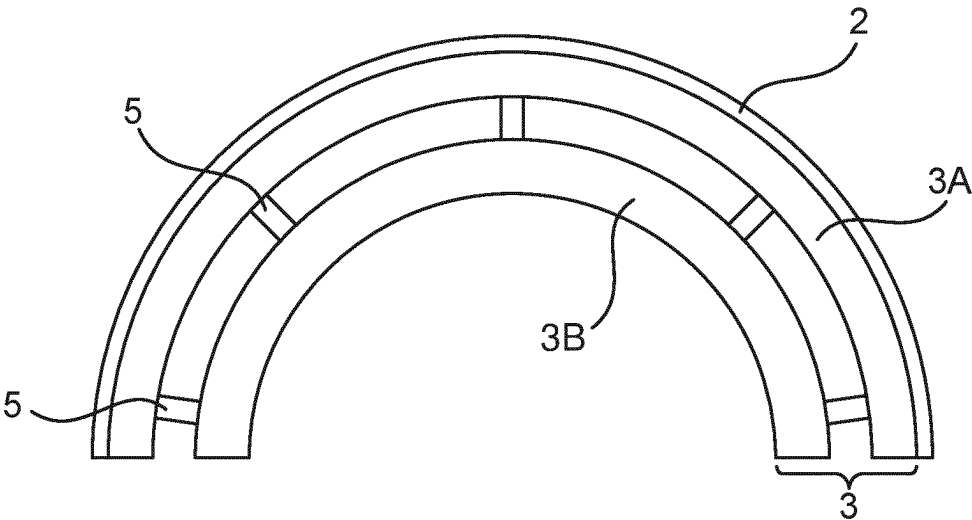


Fig. 7

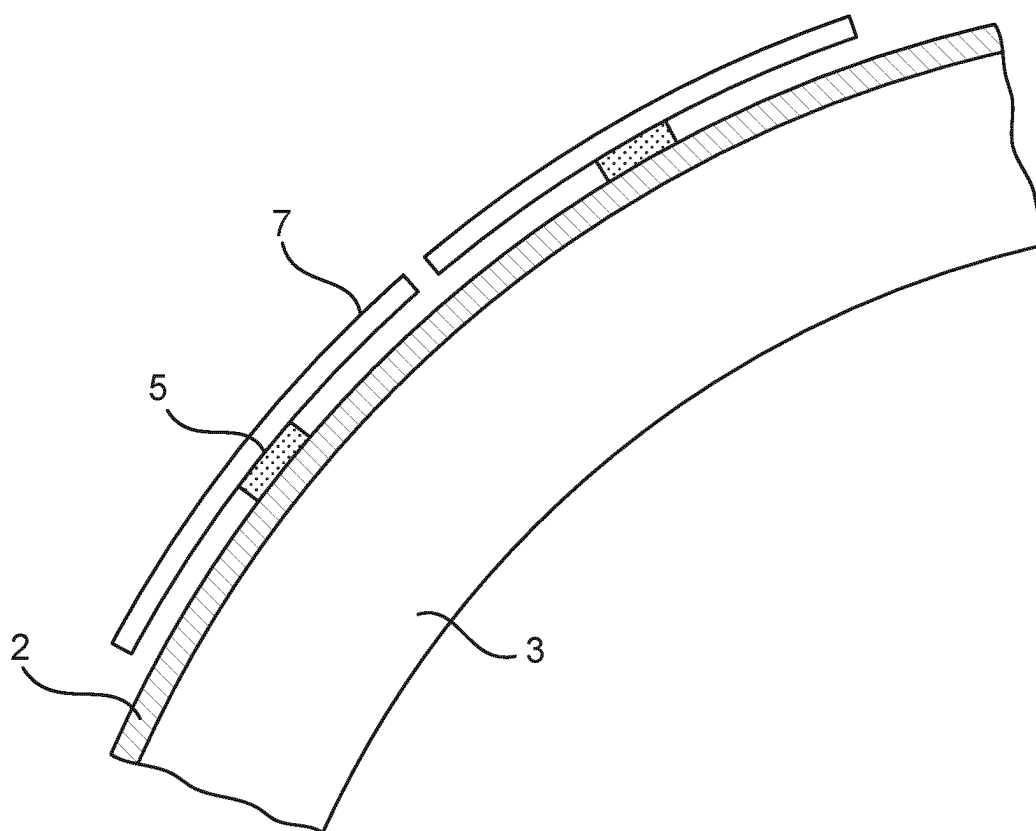


Fig. 8

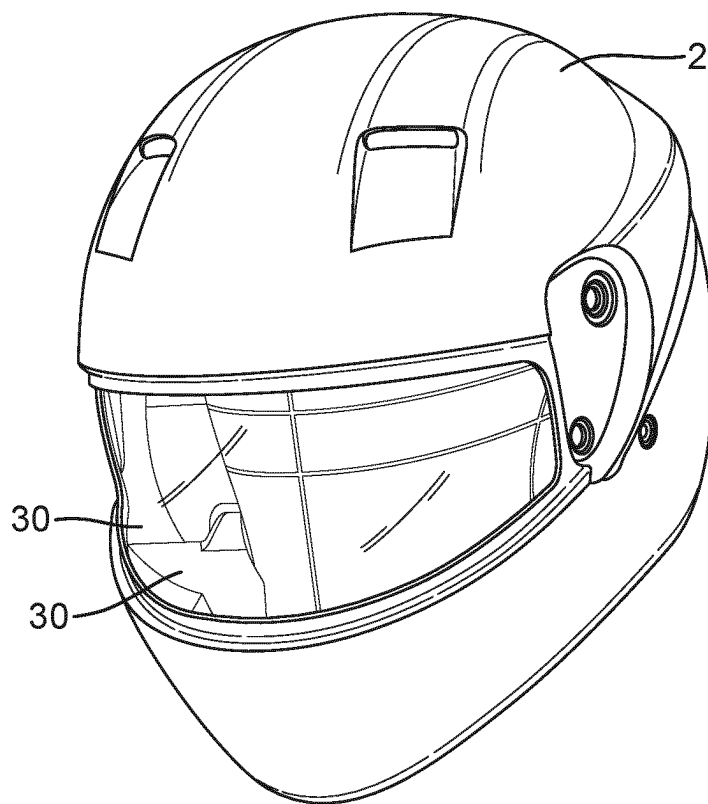


Fig. 9

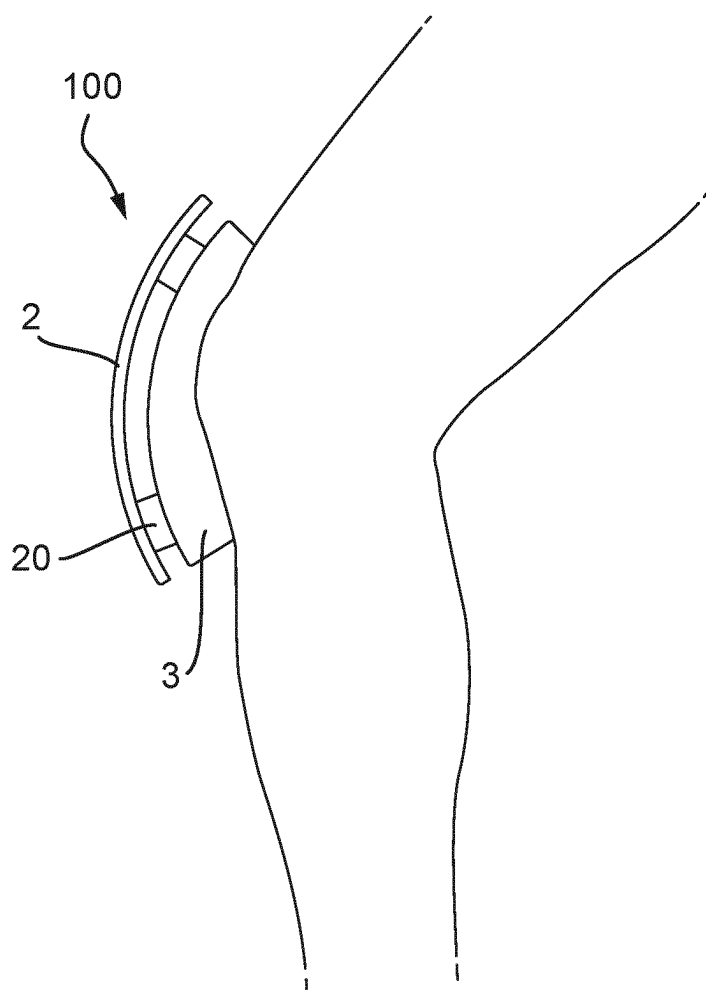


Fig. 10

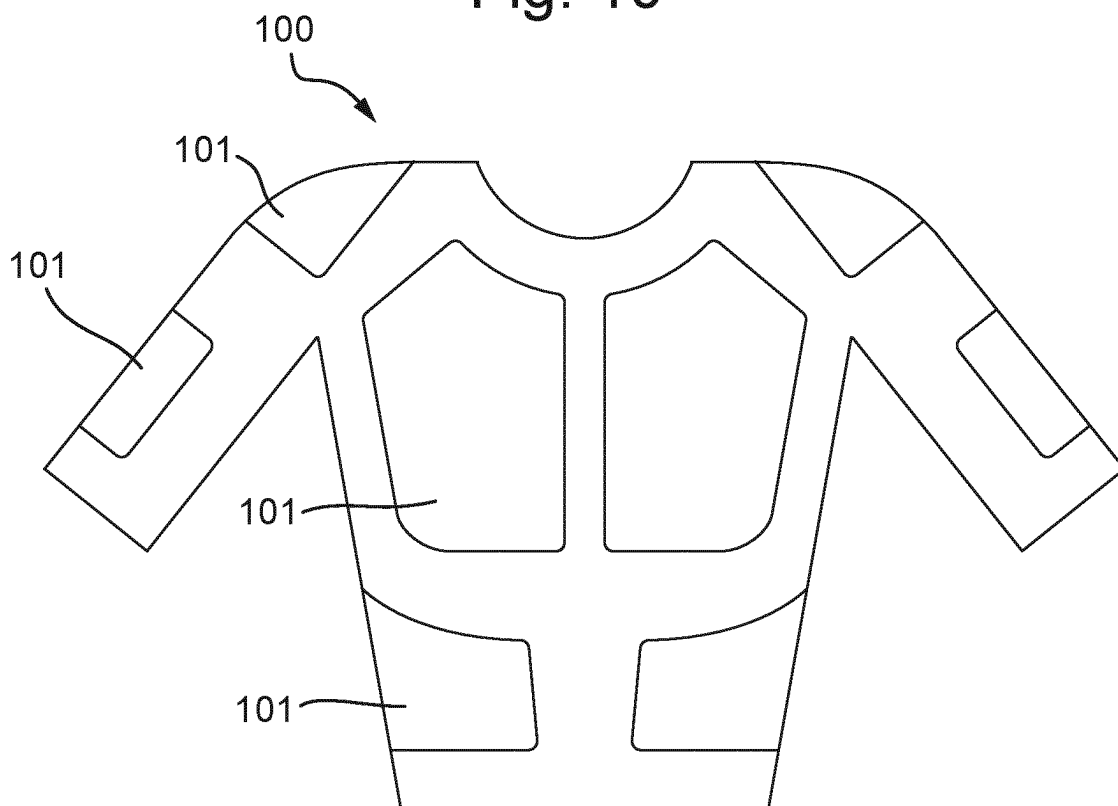


Fig. 11

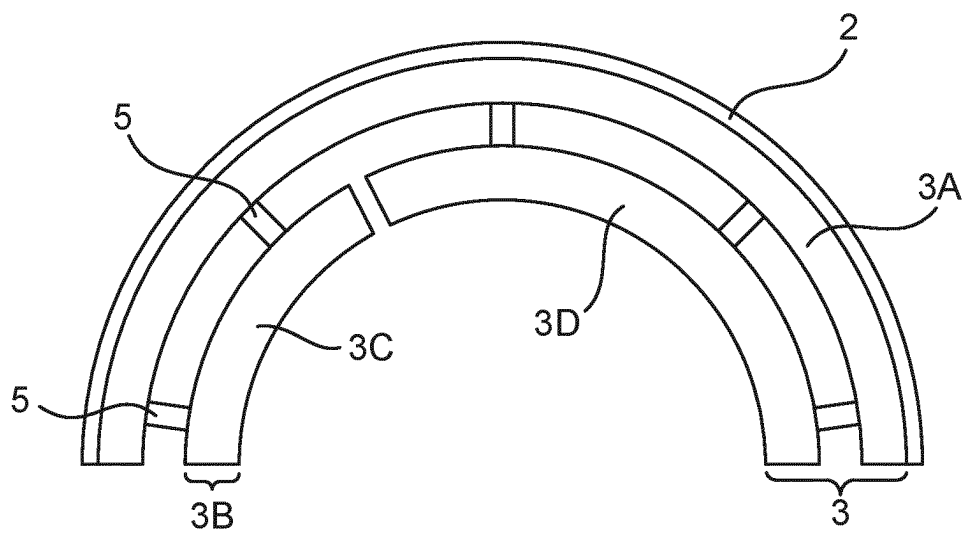


Fig. 12

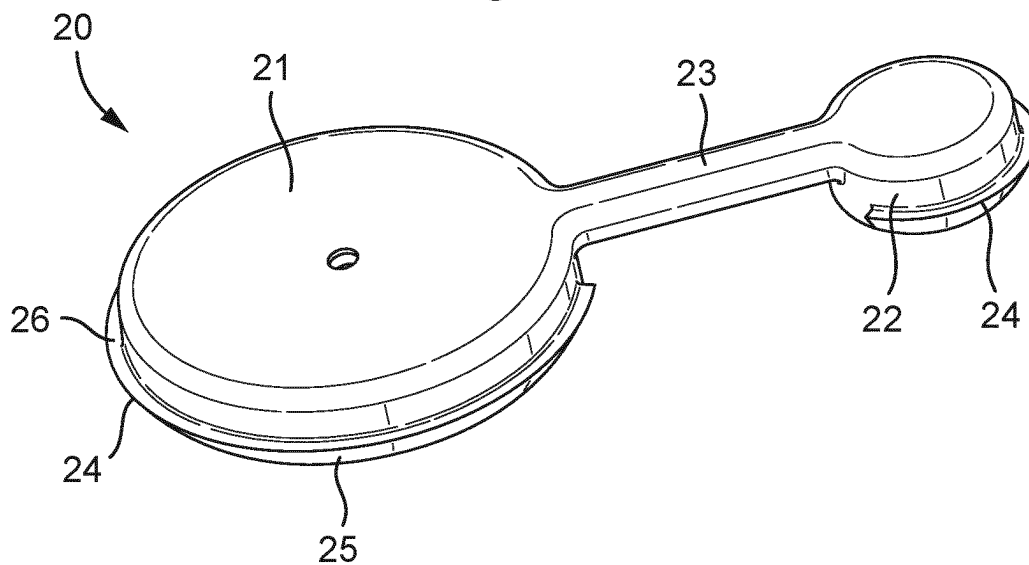


Fig. 13

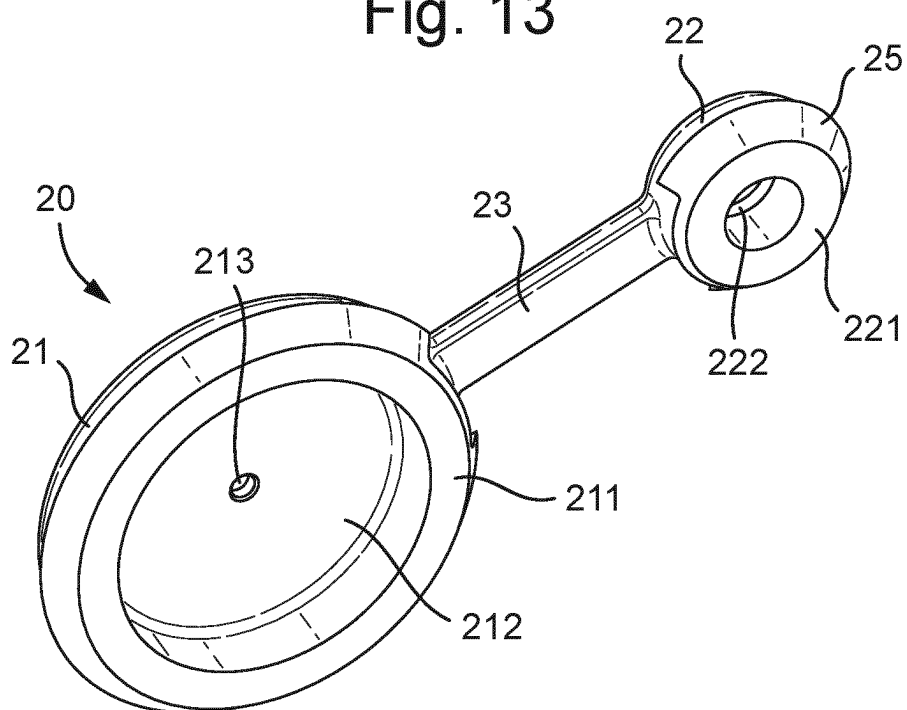


Fig. 14

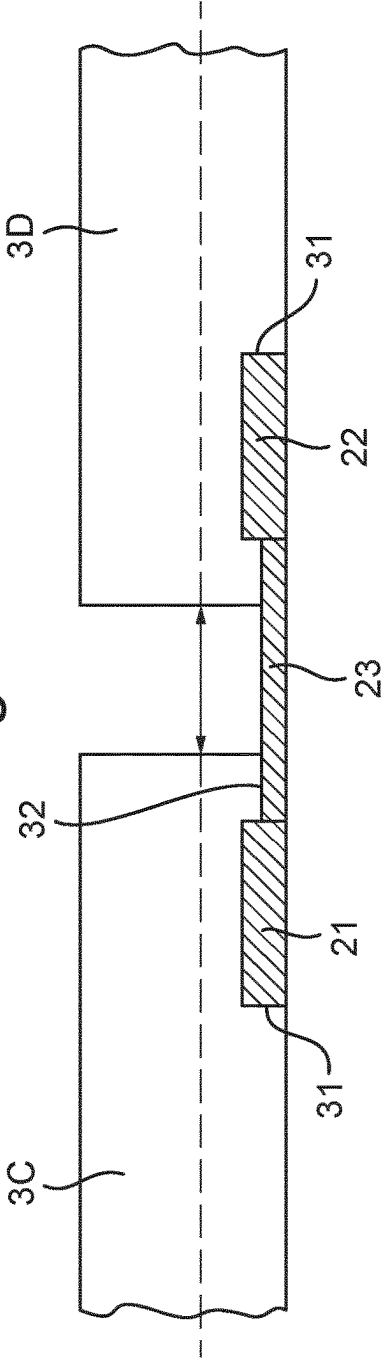


Fig. 15

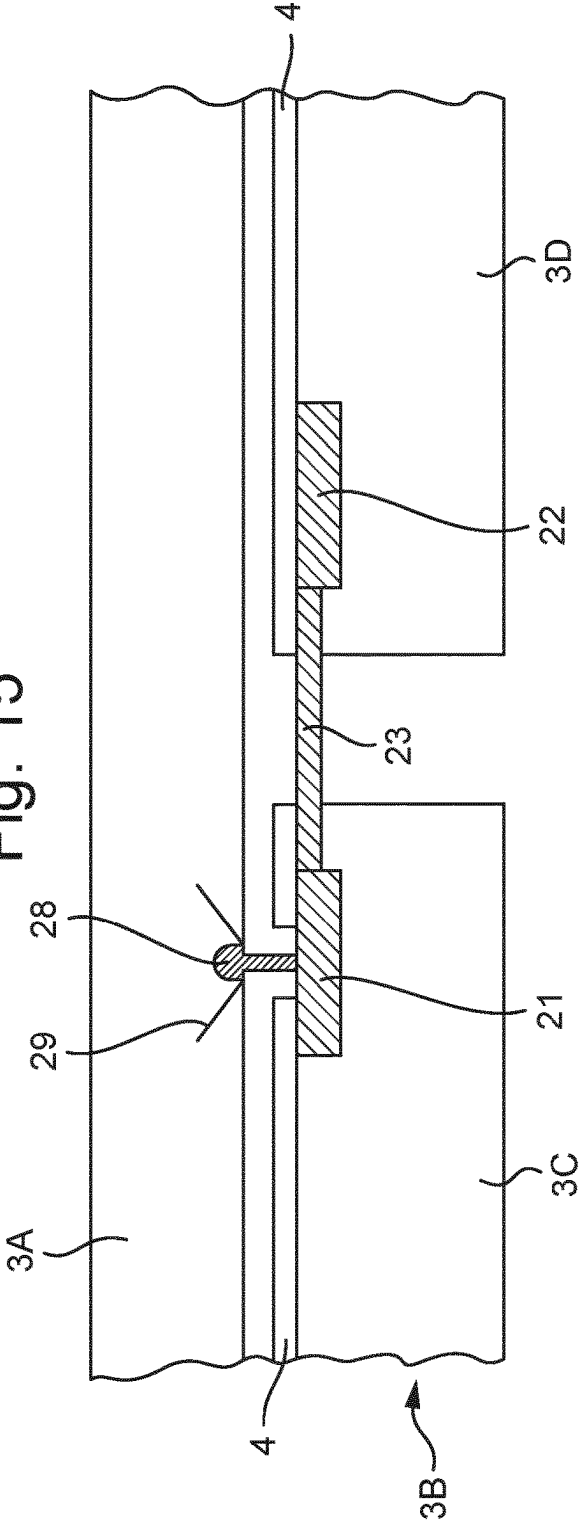


Fig. 16

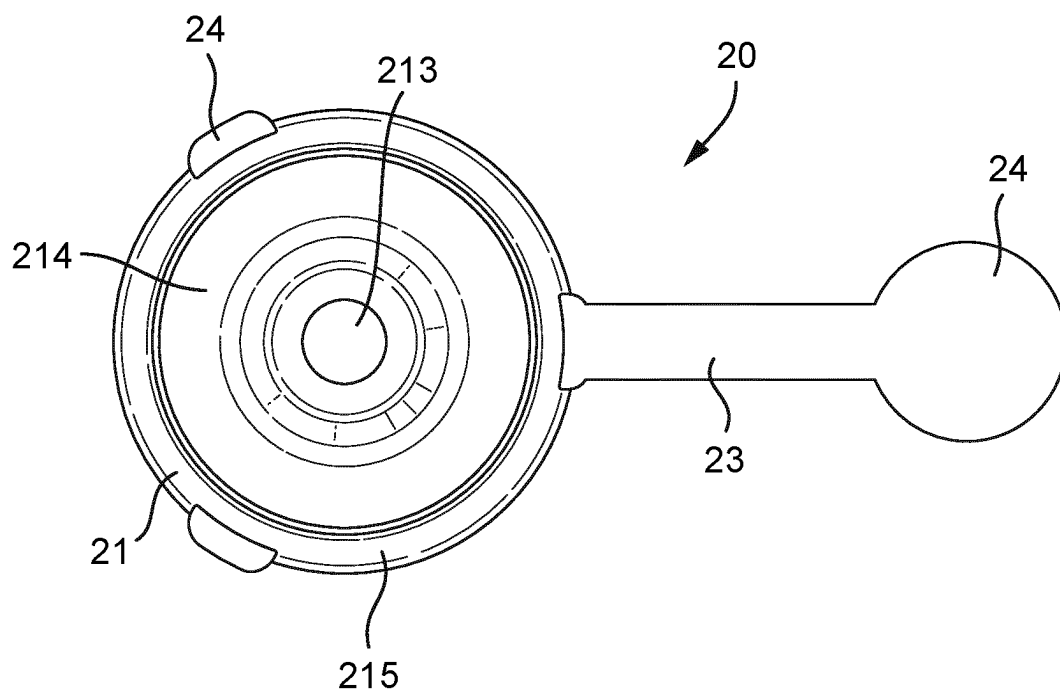


Fig. 17

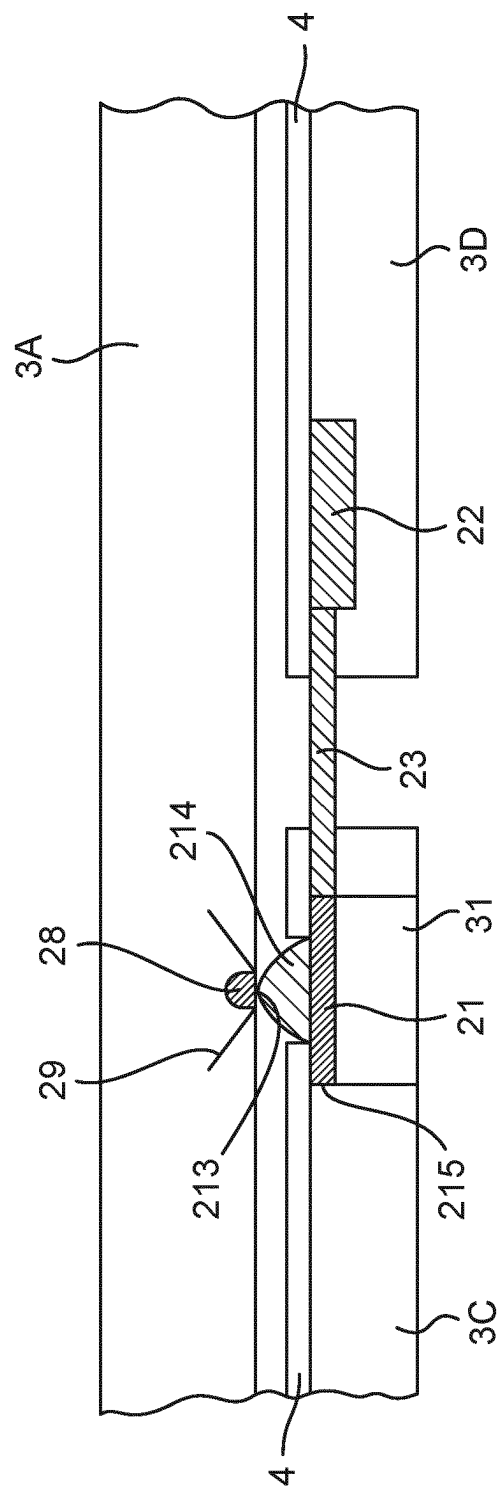


Fig. 18

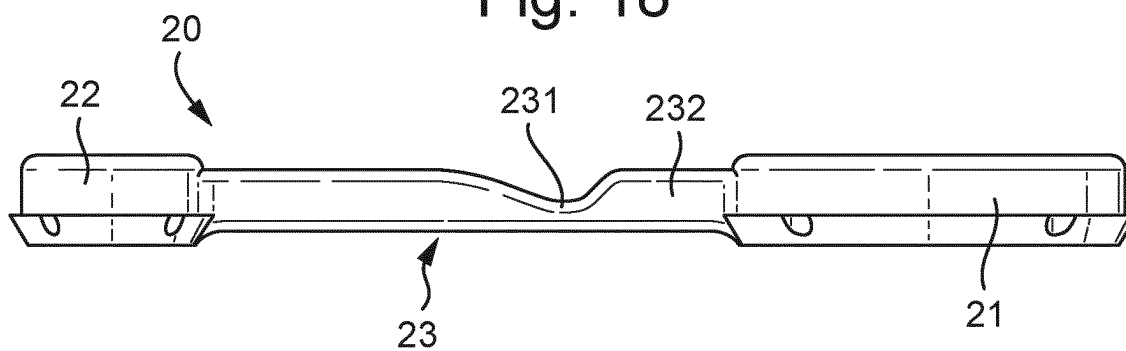


Fig. 19

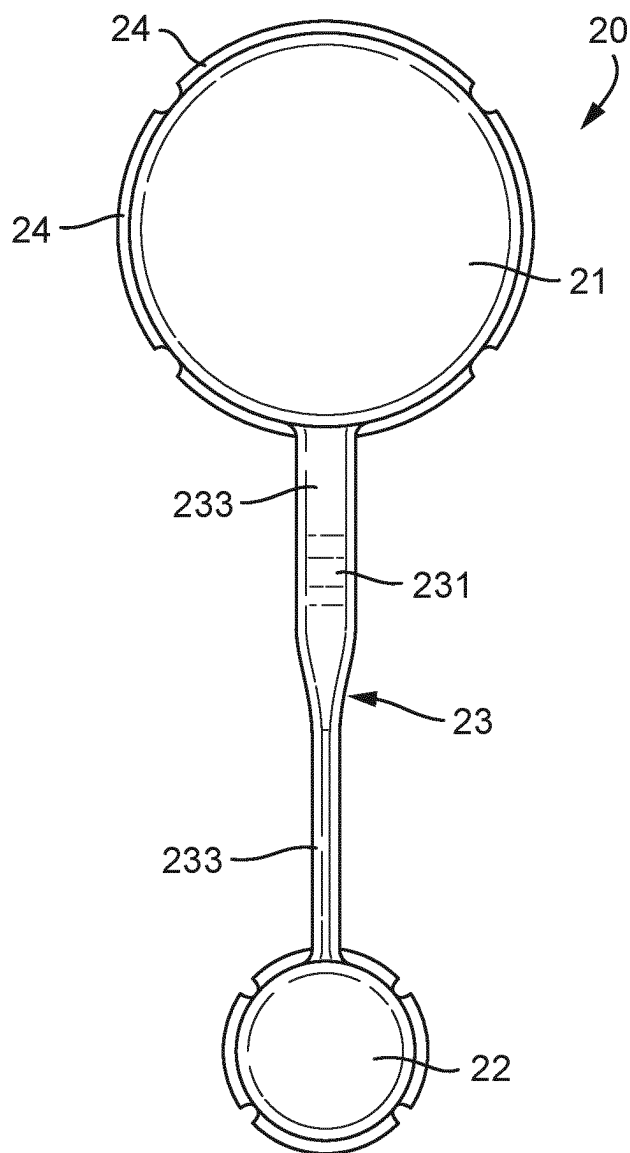


Fig. 20

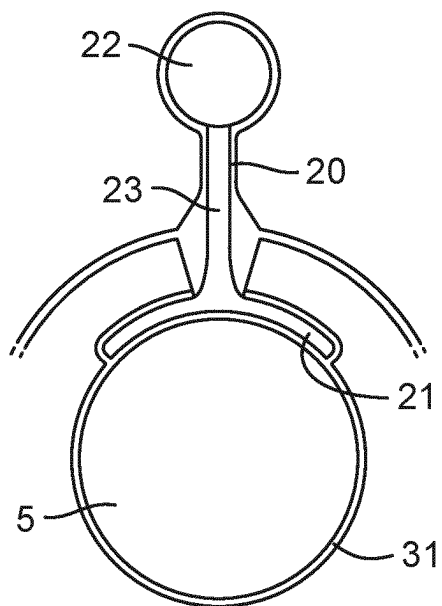


Fig. 21

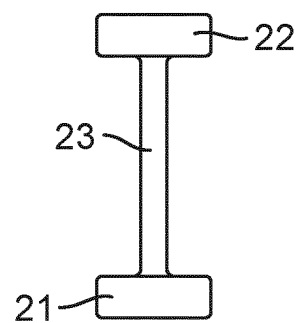


Fig. 22

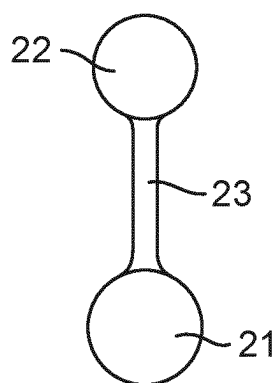


Fig. 23

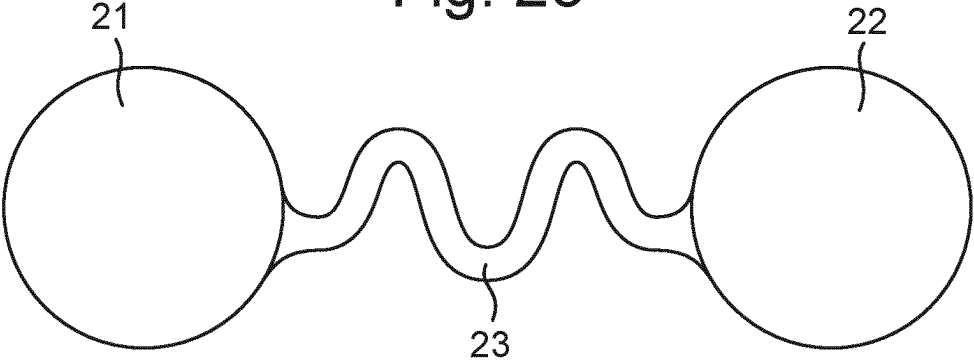


Fig. 24

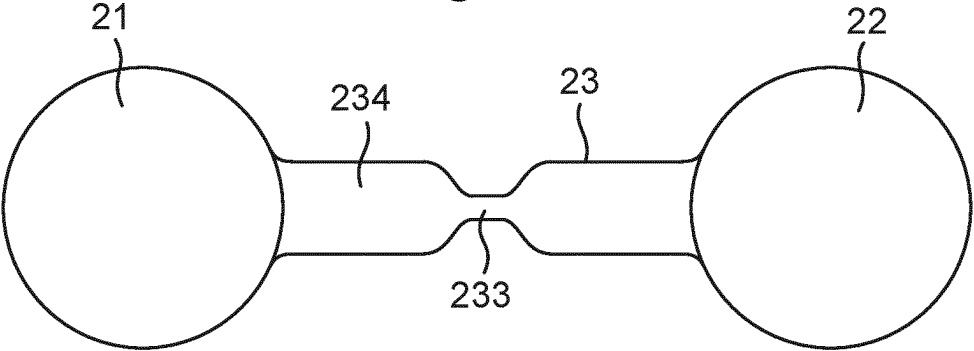


Fig. 25

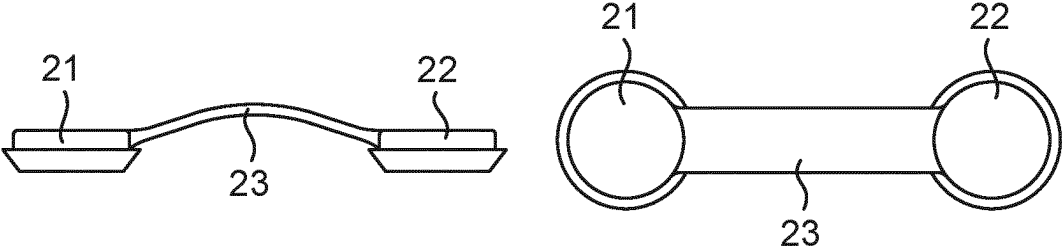


Fig. 26

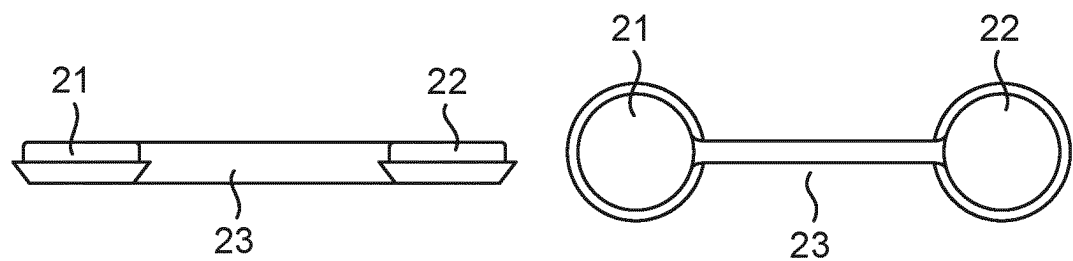


Fig. 27

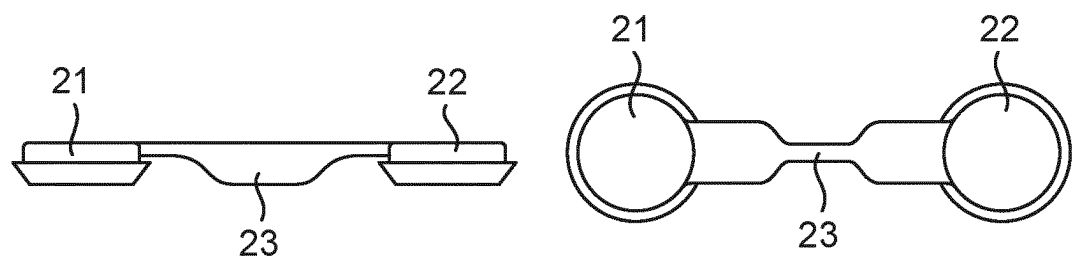


Fig. 28

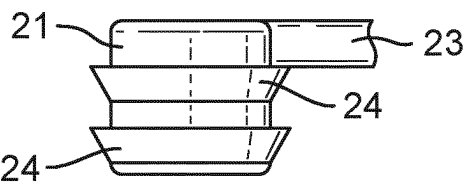


Fig. 29

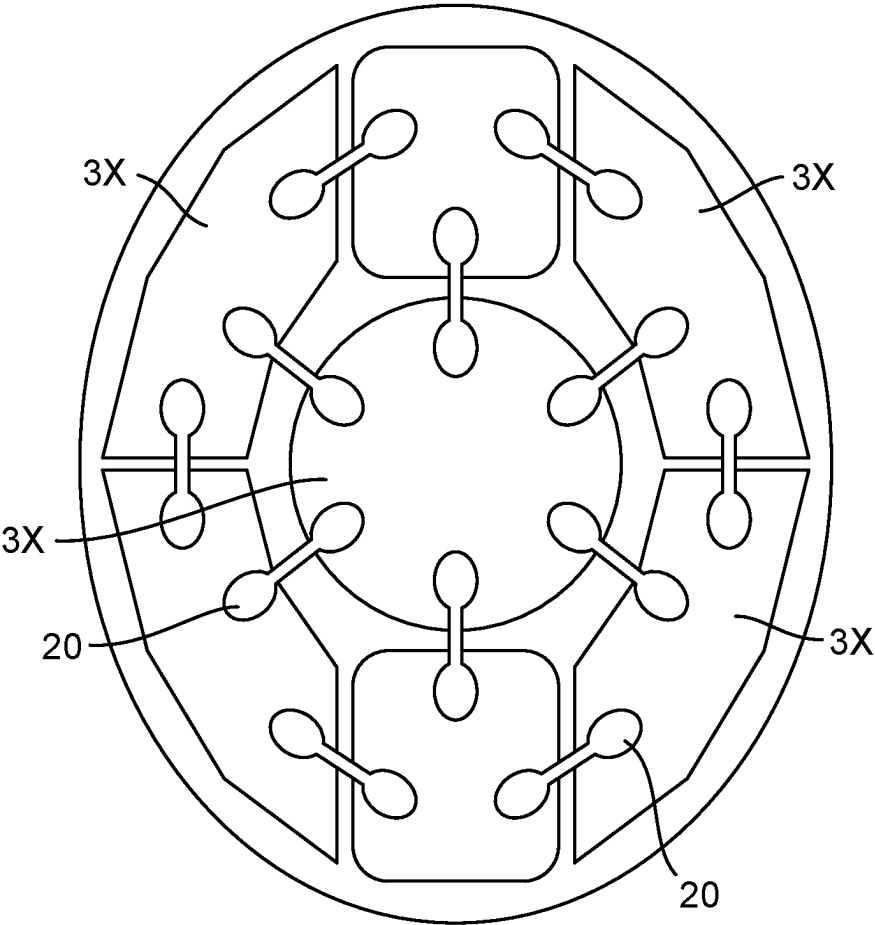


Fig. 30

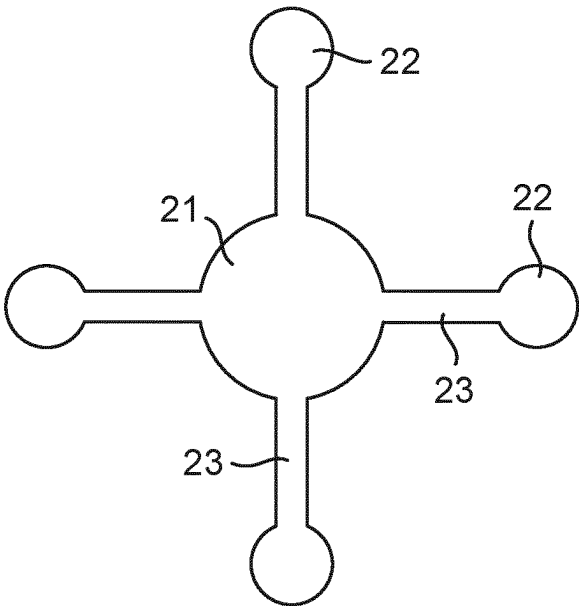


Fig. 31

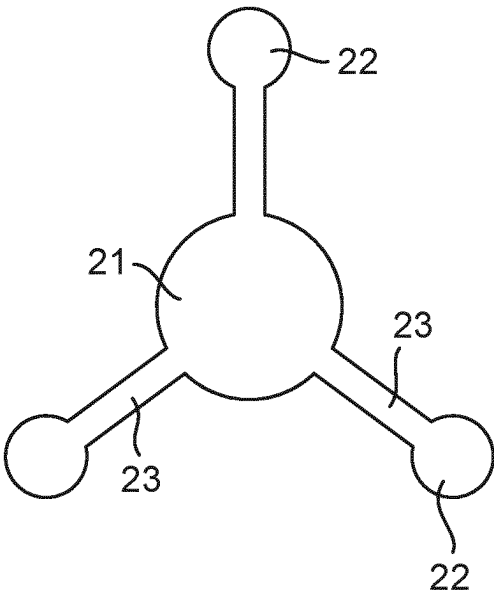


Fig. 32

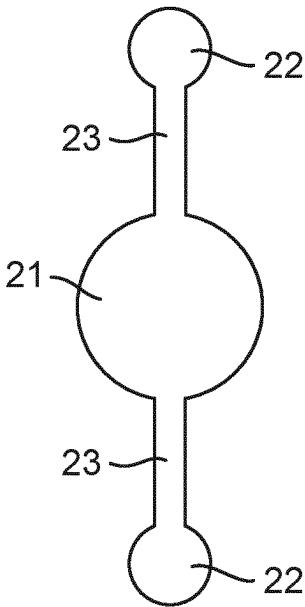
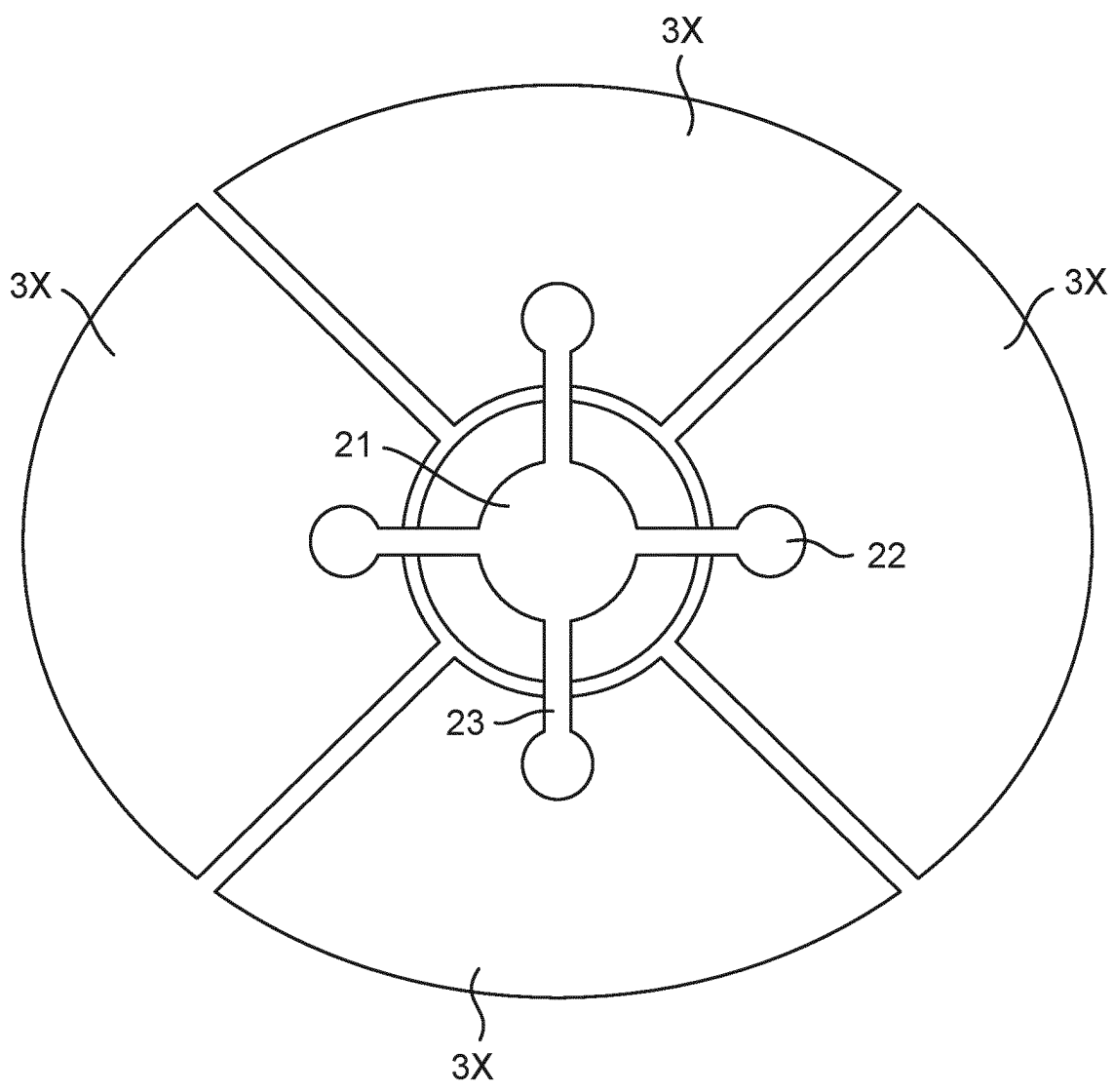


Fig. 33



CONNECTOR AND APPARATUS

TECHNICAL FIELD

[0001] The present disclosure relates to a connector for connecting parts of an apparatus and an apparatus.

BACKGROUND ART

[0002] Impact protection apparatuses generally aim to reduce the energy transferred to an object, such as a person to be protected, by an impact. This may be achieved by energy absorbing means, energy redirecting means, or a combination thereof. Energy absorbing means may include energy absorbing materials, such as a foam materials, or structures configured to deform elastically and/or plastically in response to an impact. Energy redirecting means may include structures configured to slide, shear or otherwise move in response to an impact.

[0003] Impact protection apparatuses include protective apparel for protecting a wearer of the apparel. Protective apparel comprising energy absorbing means and/or energy redirecting means is known. For example, such means are implemented extensively in protective headgear, such as helmets.

[0004] Examples of helmets comprising energy absorbing means and energy redirecting means include WO 2001/045526 and WO 2011/139224 (the entirety of which are herein incorporated by reference). Specifically, these helmets include at least one layer formed from an energy absorbing material and at least one layer that can move relative to the head of the wearer of the helmet under an impact.

[0005] Implementing moving parts in a protective apparatus has challenges. For example, connecting two layers of an apparatus in such a way that permits enough relative movement between parts of the apparatus under an impact but maintains the structural integrity of the apparatus can be challenging. Ensuring that the connector can be manufactured and assembled relatively easily can be challenging.

[0006] It is the aim of the present invention to provide a connector and an apparatus comprising the connector that at least partially address some of the problems discussed above.

[0007] Further, ensuring desired relative movement between moving parts under an impact can be challenging. Ensuring that the helmet/the apparatus can be manufactured and assembled relatively easily can be challenging.

[0008] It is the aim of the present invention to provide an apparatus that at least partially addresses some of the problems discussed above.

SUMMARY OF THE INVENTION

[0009] According to an aspect of the disclosure there is provided a connector for connecting first and second parts of an apparatus, the first and second parts being configured to move relative to each other at least in a first plane, the connector comprising: a first attachment part for connecting to the first part of the apparatus; a second attachment part for connecting to the second part of the apparatus; a resilient part extending between the first and second attachment parts and configured to deform to allow relative movement between the first and second attachment parts, wherein: the first attachment part and the second attachment part are displaced relative to each other in a first direction, and the

resilient part is narrower, in a second direction perpendicular to the first direction, than the first attachment part and the second attachment part.

[0010] Optionally, at least one of the first attachment part and the second attachment part is configured to press fit into a corresponding recess in the respective first or second part of the apparatus.

[0011] Optionally, the first attachment part and the second attachment part are displaced relative to each other in a first direction, and the connector is configured to connect the first and second parts of the apparatus such that, in use, the first direction is substantially parallel to the first plane, and the at least one of the first attachment part and the second attachment part is configured to be inserted into the corresponding recess in the respective first or second part of the apparatus in a direction substantially perpendicular to the first plane.

[0012] Optionally, the at least one of the first attachment part and the second attachment part comprises one or more protrusions configured to engage with the corresponding recess. Optionally, the one or more protrusions comprises a sloped portion and a shelf, the sloped portion being configured to ease insertion of the attachment part into the recess and the shelf configured to resist removal of the attachment part from the recess.

[0013] Optionally, the first attachment part and/or the second attachment part are formed from a resilient material. Optionally, the resilient part is formed from the same resilient material.

[0014] Optionally, at least one of the first and second attachment parts comprises an anchor point configured to connect the connector to a third part of the apparatus or a further connector. Optionally, the anchor point comprise a hole configured to receive a snap-pin.

[0015] Optionally, the first and/or second attachment part is substantially circular.

[0016] Optionally, the resilient part is elongate.

[0017] Optionally, the connector comprises multiple second attachment parts, and multiple respective resilient parts, for connecting to one or more second parts of the apparatus.

[0018] Optionally, the connector further comprise a third attachment part for connecting to a third part of the apparatus, the third part of the apparatus being configured to move relative to the first and second parts in a second plane substantially parallel to the first plane; and a second resilient part extending between the first attachment part and the third attachment part and configured to deform to allow relative movement between the third and first attachment parts.

[0019] According to a second aspect of the disclosure there is provided a connector for connecting first, second and third parts of an apparatus, the first and second parts being configured to move relative to each other at least in a first plane, and the third part being configured to move relative to the first and second parts in a second plane substantially parallel to the first plane, the connector comprising: a first attachment part for connecting to the first part of the apparatus; a second attachment part for connecting to the second part of the apparatus; a third attachment part for connecting to the third part of the apparatus; a first resilient part extending between the first and second attachment parts and configured to deform to allow relative movement between the first and second attachment parts; and a second resilient part extending between the first attachment part and

the third attachment part and configured to deform to allow relative movement between the third and first attachment parts.

[0020] Optionally, the first and third attachment parts are detachable from each other, such that the connector is formed in two parts.

[0021] Optionally, the first attachment part comprises a peripheral portion, surrounding the second resilient part about a first axis, and the second resilient part surrounds the third attachment part about a second axis, and the resilient portion is configured to protrude from the peripheral portion in a direction along the first axis, when the connector is connected to the apparatus.

[0022] According to a third aspect of the disclosure there is provided an apparatus comprising: a first part and a second part, the first and second parts being configured to move relative to each other at least in a first plane; a connector according to any preceding claim, wherein the first attachment part of the connector is connected to the first part of the apparatus and the second attachment part of the connector is connected to the second part of the apparatus.

[0023] Optionally, the first part comprises a first recess and the first attachment part is arranged in the first recess, and/or the second part comprises a second recess and the second attachment part of the connector is arranged in the second recess.

[0024] Optionally, the first part and/or the second part comprise a channel in communication with a respective recess, wherein the resilient part is arranged in the channel.

[0025] Optionally, the first and second recesses are substantially the same shape as the respective attachment parts.

[0026] Optionally, at least one of the first and second attachment parts is press fit into the respective recess.

[0027] Optionally, when the connector comprises a third attachment part, as above, and the third attachment part is also arranged in the first recess.

[0028] Optionally, the apparatus has a layered structure and the first and second parts together form a first layer.

[0029] Optionally, when the connector comprises multiple second attachment parts, as above, and the apparatus comprises multiple second parts connected by the connector.

[0030] Optionally, the apparatus further comprises a third part configured to move relative to the first and second parts in a second plane substantially parallel to the first plane. Optionally, the connector comprises a third attachment part, as above, and the third attachment part is connected to the third part of the apparatus.

[0031] Optionally, when the apparatus comprises a first layer, as above, and the third part forms a second layer adjacent the first layer.

[0032] Optionally, the first and second parts are energy absorbing parts, optionally forming a first energy absorbing layer.

[0033] Optionally, when the apparatus comprises a third part, as above, the third part is an energy absorbing part, optionally forming a second energy absorbing layer.

[0034] Optionally, the apparatus is a helmet.

BRIEF DESCRIPTION OF THE DRAWINGS

[0035] The invention is described in detail below, with reference to the accompanying figures, in which:

[0036] FIG. 1 schematically shows a cross-section through a first example helmet;

[0037] FIG. 2 schematically shows a cross-section through a second example helmet;

[0038] FIG. 3 schematically shows a cross-section through a third example helmet;

[0039] FIG. 4 schematically shows a cross-section through a fourth example helmet;

[0040] FIG. 5 schematically shows a cross-section through a fifth example helmet;

[0041] FIG. 6 schematically shows a cross-section through a sixth example helmet;

[0042] FIG. 7 schematically shows a cross-section through a seventh example helmet;

[0043] FIG. 8 shows an eighth example helmet;

[0044] FIG. 9 shows a first example of body armour;

[0045] FIG. 10 shows a second example of body armour;

[0046] FIG. 11 schematically shows a cross-section through a ninth example helmet;

[0047] FIG. 12 shows a first view of a first example connector;

[0048] FIG. 13 shows a first view of the first example connector;

[0049] FIG. 14 shows the first example connector in a first arrangement in a helmet;

[0050] FIG. 15 shows the first example connector in a second arrangement in a helmet;

[0051] FIG. 16 shows a second example connector;

[0052] FIG. 17 shows the second example connector in a first arrangement in a helmet;

[0053] FIG. 18 shows a first view of a third example connector;

[0054] FIG. 19 shows a second view of the third example connector;

[0055] FIG. 20 shows a fourth example connector;

[0056] FIG. 21 shows a fifth example connector;

[0057] FIG. 22 shows a sixth example connector;

[0058] FIG. 23 shows a seventh example connector;

[0059] FIG. 24 shows an eighth example connector;

[0060] FIG. 25 shows a ninth example connector;

[0061] FIG. 26 shows a tenth example connector;

[0062] FIG. 27 shows an eleventh example connector;

[0063] FIG. 28 shows a twelfth example connector;

[0064] FIG. 29 shows a further example helmet;

[0065] FIG. 30 shows a thirteenth example connector;

[0066] FIG. 31 shows a fourteenth example connector;

[0067] FIG. 32 shows a fifteenth example connector; and

[0068] FIG. 33 shows a further example helmet.

DETAILED DESCRIPTION

[0069] It should be noted that the Figures are schematic, the proportions of the thicknesses of the various layers, and/or of any gaps between layers, depicted in the Figures have been exaggerated for the sake of clarity and can of course be adapted according to need and requirements.

[0070] Although the examples described below relate to helmets, it should be understood that the invention applies generally to protective apparatuses, including other types of headgear and other protective apparel.

[0071] Protective apparatuses can be understood to have parts corresponding to the parts of the helmets described below. For example, protective apparatuses may have a layered structure corresponding to the layered structure of the described helmets.

[0072] Terms that are specific to a helmet, such as “radial direction” can be understood to have equivalents in the

context of other protective equipment, such as “thickness direction”. A “wearer” is to generally understood as corresponding to an object that is to be protected by the protective apparatus, and “head” as a specific part of the object, e.g. a different body part, with which the apparatus is in contact.

[0073] General features of the example helmets are described below with reference to FIGS. 1 to 7.

[0074] FIGS. 1 to 7 show example helmets 1 comprising an energy absorbing layer 3. The purpose of the energy absorbing layer 3 is to absorb and dissipate energy from an impact in order to reduce the energy transmitted to the wearer of the helmet. Within the helmet 1, the energy absorbing layer may be the primary energy absorbing element. Although other elements of the helmet 1 may absorb that energy to a more limited extent, this is not their primary purpose.

[0075] The energy absorbing layer 3 may absorb energy from a radial component of an impact more efficiently than a tangential component of an impact. The term “radial” generally refers to a direction substantially toward the centre of the wearers head, e.g. substantially perpendicular to an outer surface of the helmet 1. The term “tangential” may refer to a direction substantially perpendicular to the radial direction, in a plane comprising the radial direction and the impact direction.

[0076] The energy absorbing layer may be formed from an energy absorbing material, such as a foam material. Preferable such materials include expanded polystyrene (EPS), expanded polypropylene (EPP), expanded polyurethane (EPU), vinyl nitrile foam; or strain rate sensitive foams such as those marketed under the brand-names Poron™ and D3O™.

[0077] Alternatively, or additionally, the energy absorbing layer may have a structure that provides energy absorbing characteristics. For example, the energy absorbing layer may comprise deformable elements, such as cells or finger-like projections, that deform upon impact to absorb and dissipate the energy of an impact.

[0078] As illustrated in FIG. 6, the energy absorbing layer 3 of the helmet 1 may be divided into outer and inner parts 3A, 3B.

[0079] As illustrated in FIG. 11, the energy absorbing layer 3 may be divided into multiple parts arranged adjacent each other in the circumferential direction of the helmet. FIG. 11 shows such a helmet of the type shown in FIG. 6, with the inner parts 3B being formed in front and back parts 3C and 3D.

[0080] The energy absorbing layer is not limited to one specific arrangement or material. The energy absorbing layer 3 may be provided by multiple layers having different arrangements, i.e. formed from different materials or having different structures. The energy absorbing layer 3 may be a relatively thick layer. For example, it may be thickest layer of the helmet 1.

[0081] FIGS. 1 to 7 show example helmets 1 comprising an outer layer 2. The purpose of the outer layer 2 may be to provide rigidity to the helmet. This may help spread the impact energy over a larger area of the helmet 1. The outer layer 2 may also provide protection against objects that might pierce the helmet 1. Accordingly, the outer shell may be a relatively strong and/or rigid layer, e.g. compared to an energy absorbing layer 3. The outer layer 2 may be a relatively thin layer, e.g. compared to an energy absorbing layer 3.

[0082] The outer layer 2 may be formed from a relatively strong and/or rigid material. Preferable such materials include a polymer material such as polycarbonate (PC), polyvinylchloride (PVC) or acrylonitrile butadiene styrene (ABS) for example. Advantageously, the polymer material may be fibre-reinforced, using materials such as glass-fibre, Aramid, Twaron, carbon-fibre and/or Kevlar.

[0083] As shown in FIG. 7, one or more outer plates 7 may be mounted to the outer layer 2 of the helmet 1. The outer plates 7 may be formed from a relatively strong and/or rigid material, for example from the same types of materials as from which the outer layer 2 may be formed. The selection of material used to form the outer plates 7 may be the same as, or different from, the material used to form the outer layer 2.

[0084] In some example helmets, the outer layer 2 and/or the energy absorbing layer 3 may be adjustable in size in order to provide a customised fit. For example the outer layer 2 may be provided in separate front and back parts. The relative position of the front and back parts may be adjusted to change the size of the outer layer 2. In order to avoid gaps in the outer layer 2, the front and back parts may overlap. The energy absorbing layer 3 may also be provided in separate front and back parts. These may be arranged such that the relative position of the front and back parts may be adjusted to change the size of the energy absorbing layer 3. In order to avoid gaps in the energy absorbing layer 3, the front and back parts may overlap.

[0085] FIGS. 1 to 4 shows example helmets 1 comprising an interface layer 4. Although not shown in FIGS. 5 to 7, these example helmets may also comprise an interface layer 4. The purpose of interface layer 4 may be to provide an interface between the helmet and the wearer. In some arrangements, this may improve the comfort of the wearer. The interface layer 4 may be provided to mount the helmet on the head of a wearer. The interface layer 4 may be provided as a single part or in multiple sections.

[0086] The interface layer 4 may be configured to at least partially conform to the head of the wearer. For example, the interface layer 4 may be elastically and/or may comprise an adjustment mechanism for adjusting the size of the interface layer 4. In an arrangement, the interface layer may engage with the top of a wearer’s head. Alternatively or additionally, the interface layer 4 may comprise an adjustable band configured to encircle the wearer’s head.

[0087] The interface layer 4 may comprise comfort padding 4A. Multiple sections of comfort padding 4A may be provided. The comfort padding 4A may be provided on a substrate 4B for mounting the comfort padding to the rest of the helmet 1.

[0088] The purpose of the comfort padding 4A is to improve comfort of wearing the helmet and/or to provide a better fit. The comfort padding may be formed from a relatively soft material, e.g. compared to the energy absorbing layer 3 and/or the outer layer 2. The comfort padding 4A may be formed from a foam material. However, the foam material may be of lower density and/or thinner than foam materials used for the energy absorbing layer 3. Accordingly, the comfort padding 4A will not absorb a meaningful amount of energy during an impact, i.e. for the purposes of reducing the harm to the wearer of the helmet. Comfort padding is well recognised in the art as being distinct from energy absorbing layers, even if they may be constructed from somewhat similar materials.

[0089] The interface layer 4, and/or comfort padding 4A that may be part of it, may be removable. This may enable the interface layer 4 and/or comfort padding 4A to be cleaned and/or may enable the provision of an interface layer and/or comfort padding 4A that is configured to fit a specific wearer.

[0090] Straps, e.g. chin straps, may be provided to secure the helmet 1 to the head of the wearer.

[0091] The helmets of FIGS. 1 to 4 are configured such that the interface layer 4 is able to move, for example slide, in a tangential direction relative to the energy absorbing layer 3 in response to an impact. As shown in FIGS. 1 to 4, the helmet may also comprise connectors 5 between the energy absorbing layer 3 and the interface layer 4 that allow relative movement between the energy absorbing layer 3 and the interface layer 4 while connecting the elements of the helmet together.

[0092] The helmet of FIG. 5 is configured such that the outer layer 2 is able to move, for example slide, in a tangential direction relative to the energy absorbing layer 3 in response to an impact. As shown in FIG. 5, the helmet 1 may also comprise connectors 5 between the energy absorbing layer 3 and the outer layer 2 that allow relative movement between the energy absorbing layer 3 and the outer layer 2 while connecting the elements of the helmet together.

[0093] The helmet of FIG. 6 is configured such that the outer part 3A of the energy absorbing layer 3 is able to move, for example slide, in a tangential direction relative to the inner part 3B of the energy absorbing layer 3 in response to an impact. As shown in FIG. 6, the helmet 1 may also comprise connectors 5 between the outer part 3A of the energy absorbing layer 3 and the inner part 3B of the energy absorbing layer 3, that allow relative movement between the outer part 3A of the energy absorbing layer 3 and the inner part 3B of the energy absorbing layer 3, while connecting the elements of the helmet together.

[0094] In examples, such as that of FIG. 11, in which an energy absorbing layer is split into multiple parts 3C and 3D arranged adjacent each other in the circumferential direction of the helmet, these parts may be configured to move relative to each other, as well as other parts of the helmet.

[0095] The helmet of FIG. 7 is configured such that the outer plates 8 are able to move, for example slide, in a tangential direction relative to the outer layer 2 in response to an impact. As shown in FIG. 7, the helmet may also comprise connectors 5 between the outer plates 8 and the outer layer 2 that allow relative movement between the outer plates 7 and the outer layer 2, while connecting the elements of the helmet together.

[0096] The purpose of helmet layers that move or slide relative to each other may be to redirect energy of an impact that would otherwise be transferred to the head the wearer. This may improve the protection afforded to the wearer against a tangential component of the impact energy. A tangential component of the impact energy would normally result in rotational acceleration of the head of the wearer. It is well known that such rotation can cause brain injury. It has been shown that helmets with layers that move relative to each other can reduce the rotational acceleration of the head of the wearer. A typical reduction may be roughly 25% but reductions as high as 90% may be possible in some instances.

[0097] Preferably, relative movement between helmet layers results in a total shift amount of at least 0.5 cm between

an outermost helmet layer and an inner most helmet layer, more preferably at least 1 cm, more preferably still at least 1.5 cm. Preferably the relative movement can occur in any direction, e.g. in a circumferential direction around the helmet, left to right, front to back and any direction in between.

[0098] Relative movement can be considered to occur substantially in a plane over the relevant ranges, even though movement between layers may be rotational rather than linear. Accordingly, reference may be made below to movement in a plane.

[0099] Regardless of how helmet layers are configured to move relative to each other, it is preferable that the relative movement, such as sliding, is able to occur under forces typical of an impact for which the helmet is designed (for example an impact that is expected to be survivable for the wearer). Such forces are significantly higher than forces that a helmet may be subject to during normal use. Impact forces tend to compress layers of the helmet together, increasing the reaction force between components and thus increasing frictional forces. Where helmets are configured to have layers sliding relative to each other the interface between them may need to be configured to enable sliding even under the effect of the high reaction forces experienced between them under an impact.

[0100] As shown in FIGS. 1 to 7, a sliding interface may be provided between the layers of the helmet 1 that are configured to slide relative to each other. At the sliding interface, surfaces slide against each other to enable relative sliding between the layers of the helmet 1. The sliding interface may be a low friction interface. Accordingly, friction reducing means may be provided at the sliding interface. Example sliding interfaces are described further below, in relation to each of the example helmets 1 shown in FIGS. 1 to 7.

[0101] The friction reducing means may be a low friction material or lubricating material. These may be provided as a continuous layer, or multiple discrete patches, or portions of material, for example. Possible low friction materials for the friction reducing means include waxy polymers such as PC, PTFE, ABS, PVC, Nylon, PFA, FEP, PE and UHMWPE, Teflon™, a woven fabric such as Tamarack™, a non-woven fabric, such a felt. Such low friction materials may have a thickness of roughly 0.1-5 mm, but other thicknesses can also be used, depending on the material selected and the performance desired. Possible lubricating materials include oils, polymers, microspheres, or powders. Combinations of the above may be used.

[0102] In one example the low friction material or lubricating material may be a polysiloxane-containing material. In particular the material may comprise (i) an organic polymer, a polysiloxane and a surfactant; (ii) an organic polymer and a copolymer based on a polysiloxane and an organic polymer; or (iii) a non-elastomeric cross-linked polymer obtained or obtainable by subjecting a polysiloxane and an organic polymer to a cross-linking reaction. Preferred options for such materials are described in WO2017148958.

[0103] In one example the low friction material or lubricating material may comprise a mixture of (i) an olefin polymer, (ii) a lubricant, and optionally one or more further agents. Preferred options for such materials are described in WO2020115063.

[0104] In one example the low friction material or lubricating material may comprise an ultra high molecular weight

(UHMW) polymer having a density of $\leq 960 \text{ kg/m}^3$, which UHMW polymer is preferably an olefin polymer. Preferred options for such materials are described in WO2020/115063.

[0105] In one example the low friction material or lubricating material may comprise a polyketone. Preferred options for such materials are described in WO 2020/260185.

[0106] In some arrangements, it may be desirable to configure the low friction interface such that the static and/or dynamic coefficient of friction between materials forming sliding surfaces at the sliding interface is between 0.001 and 0.3 and/or below 0.15. The coefficient of friction can be tested by standard means, such as standard test method ASTM D1894.

[0107] The friction reducing means may be provided on or be an integral part of one or both of the layers of the helmet 1 that are configured to slide relative to each other. In some examples, helmet layers may have a dual function, including functioning as a friction reducing means. Alternatively, or additionally, the friction reducing means may be a separate from the layers of the helmet 1 that are configured to slide relative to each other, but provided between the layers.

[0108] Instead of the sliding interface, in some examples, a shearing interface may be provided between the layers of the helmet 1 that are configured to move relative to each other. At the shearing interface, a shearing layer shears to enable relative movement between the layers of the helmet 1. The shearing layer may comprise a gel or liquid, which may be retained within a flexible envelope. Alternatively, the shearing layer may comprise two opposing layers connected by deformable elements that deform to enable shearing between the two opposing layers.

[0109] A single shearing layer may be provided that substantially fills the volume between two layers of a helmet. Alternatively, one or more shearing layers may be provided that fill only a portion of the volume between two layers of a helmet, e.g. leaving substantial space around the shearing layers. The space may comprise a sliding interface, as described above. As such, helmets may have a combination of shearing and sliding interfaces. Such shearing layers may act as connectors 5, which are described further below.

[0110] FIGS. 1 to 7 schematically show connectors 5. The connectors 5 are configured to connect two layers of the helmet while enabling relative movement, e.g. sliding or shearing, between the layers. Different numbers of connectors 5 may be provided than as shown in FIGS. 1 to 7. The connectors 5 may be located at different positions than as shown in FIGS. 1 to 7, for example at a peripheral edge of the helmet 1 instead of a central portion.

[0111] Typically, a connector 5 comprises first and second attachment parts respectively configured to attach to first and second parts of the helmet and a deformable part between the first and second attachment parts that enables the first and second attachment parts to move relative to each other to enable movement between the first and second parts of the helmet of the helmet. Connectors 5 may absorb some impact energy by deforming.

[0112] The specific arrangements of each of the example helmets shown in FIGS. 1 to 7 are described below.

[0113] FIG. 1 shows a helmet comprising an outer layer 2, an energy absorbing layer 3 and an interface layer 4. The interface layer 4 is provided as a single layer and comprises comfort padding.

[0114] The helmet of FIG. 1 is configured such that the interface layer 4 is able to slide relative to the energy absorbing layer 3 in response to an impact. A sliding interface is provided between the interface layer 4 and the energy absorbing layer 3.

[0115] A sliding layer 7 is provided on a surface of the energy absorbing layer 3 facing the sliding interface. The sliding layer 7 may be moulded to the energy absorbing layer 3 or otherwise attached thereto. The sliding layer 7 may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3. The sliding layer 7 is configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the sliding layer 7 from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, EEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the sliding layer 7, and/or applying a lubricant to the sliding layer 7.

[0116] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the energy absorbing layer 3 from a low friction material, by applying a low friction coating to the energy absorbing layer 3 and/or applying a lubricant to the energy absorbing layer 3.

[0117] The helmet 1 shown in FIG. 1 also comprises connectors 5 attached to the interface layer 4. The connectors are also connected to the sliding layer 7 to allow relative sliding between the energy absorbing layer 3 and the interface layer 4. Alternatively, or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as the energy absorbing layer 3 or the outer shell 2. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0118] It should be understood that such an arrangement of the energy absorbing layer 3 and the interface layer 4 may be added to any helmet described herein.

[0119] FIG. 2 shows a helmet comprising an outer layer 2, an energy absorbing layer 3 and an interface layer 4. The interface layer 4 is provided as a plurality of independent sections each comprising comfort padding.

[0120] The helmet of FIG. 2 is configured such that the section of the interface layer 4 are able to slide relative to the energy absorbing layer 3 in response to an impact. A sliding interface is provided between the sections of the interface layer 4 and the energy absorbing layer 3.

[0121] A sliding layer 7 is provided on a surface of the energy absorbing layer 3 facing the sliding interface. The sliding layer 7 may be moulded to the energy absorbing layer 3 or otherwise attached thereto. The sliding layer 7 may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3. The sliding layer 7 is configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the sliding layer 7 from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, EEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the sliding layer 7, and/or applying a lubricant to the sliding layer 7.

[0122] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the energy absorbing layer 3 from a

low friction material, by applying a low friction coating to the energy absorbing layer 3 and/or applying a lubricant to the energy absorbing layer 3.

[0123] The helmet 1 shown in FIG. 2 also comprises connectors 5 attached to each independent section of the interface layer 4. The connectors 5 are also attached to the sliding layer 7 to allow relative sliding between the energy absorbing layer 3 and the sections of the interface layer 4. Alternatively or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as the energy absorbing layer 3 or the outer shell 2. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0124] It should be understood that such an arrangement of the energy absorbing layer 3 and the interface layer 4 may be added to any helmet described herein.

[0125] FIG. 3 shows a helmet comprising an outer layer 2, an energy absorbing layer 3 and an interface layer 4. The interface layer 4 is provided as a single layer and comprises comfort padding 4A attached to a substrate 4B. The substrate 4B may be bonded to the outer side of the comfort padding 4A. Such bonding could be through any means, such as by adhesive or by high frequency welding or stitching.

[0126] The helmet of FIG. 3 is configured such that the interface layer 4 is able to slide relative to the energy absorbing layer 3 in response to an impact. A sliding interface is provided between the interface layer 4 and the energy absorbing layer 3.

[0127] The substrate 4B of the interface layer 4 faces the sliding interface. The substrate 4B may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3 and/or the comfort padding 4A. The substrate 4B is configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the substrate 4B from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, EEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the substrate 4B, and/or applying a lubricant to the substrate 4B. In alternative example, the substrate 4B may be formed from a fabric material, optionally coated with a low friction material.

[0128] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the energy absorbing layer 3 from a low friction material, by applying a low friction coating to the energy absorbing layer 3 and/or applying a lubricant to the energy absorbing layer 3.

[0129] The helmet 1 shown in FIG. 3 also comprises connectors 5 attached to the interface layer 4. The connectors are also connected to the energy absorbing layer to allow relative sliding between the energy absorbing layer 3 and the interface layer 4. Alternatively, or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as the outer shell 2. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0130] It should be understood that such an arrangement of the energy absorbing layer 3 and the interface layer 4 may be added to any helmet described herein.

[0131] FIG. 4 shows a helmet comprising an outer layer 2, an energy absorbing layer 3 and an interface layer 4. The interface layer 4 is provided as a plurality of independent sections each comprising comfort padding 4A attached to a

substrate 4B. The substrate 4B may be bonded to the outer side of the comfort padding 4A. Such bonding could be through any means, such as by adhesive or by high frequency welding or stitching.

[0132] The helmet of FIG. 4 is configured such that the interface layer 4 is able to slide relative to the energy absorbing layer 3 in response to an impact. A sliding interface is provided between the interface layer 4 and the energy absorbing layer 3.

[0133] The substrate 4B of the sections of the interface layer 4 faces the sliding interface. The substrate 4B may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3 and/or the comfort padding 4A. The substrate 4B is configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the substrate 4B from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, EEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the substrate 4B, and/or applying a lubricant to the substrate 4B. In alternative example, the substrate 4B may be formed from a fabric material, optionally coated with a low friction material.

[0134] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the energy absorbing layer 3 from a low friction material, by applying a low friction coating to the energy absorbing layer 3 and/or applying a lubricant to the energy absorbing layer 3.

[0135] The helmet 1 shown in FIG. 4 also comprises connectors 5 attached to the sections of the interface layer 4. The connectors 5 are also connected to the energy absorbing layer 3 to allow relative sliding between the energy absorbing layer 3 and the interface layer 4. Alternatively, or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as the outer shell 2. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0136] It should be understood that such an arrangement of the energy absorbing layer 3 and the interface layer 4 may be added to any helmet described herein.

[0137] FIG. 5 shows a helmet comprising an outer layer 2 and an energy absorbing layer 3. Although not shown, an interface layer may additionally be provided.

[0138] The helmet of FIG. 5 is configured such that the outer layer 2 is able to slide relative to the energy absorbing layer 3 in response to an impact. A sliding interface may be provided between the outer layer 2 and the energy absorbing layer 3.

[0139] Although not shown, an additional layer may be provided on a surface of the energy absorbing layer 3 facing the sliding interface. The additional layer may be moulded to the energy absorbing layer 3 or otherwise attached thereto. The additional layer may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3. The additional layer may be configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the additional layer from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, EEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the additional layer and/or applying a lubricant to the additional layer.

[0140] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the outer layer 2 from a low friction material, providing an additional low friction layer on a surface of the outer layer 2 facing the sliding interface, by applying a low friction coating to the outer layer 2 and/or applying a lubricant to the outer layer 2.

[0141] The helmet 1 shown in FIG. 5 also comprises connectors 5 attached to the outer layer 2. The connectors 5 are also attached to the energy absorbing layer 3 (or additional layer) to allow relative sliding between the energy absorbing layer 3 and the sections of the interface layer 4. Alternatively or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as an interface layer. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0142] It should be understood that such an arrangement of the outer shell 2 and the energy absorbing layer 3 may be added to any helmet described herein.

[0143] FIG. 6 shows a helmet comprising an outer layer 2 and an energy absorbing layer 3. As illustrated, the energy absorbing layer 3 of the helmet shown in FIG. 6 is divided into outer and inner parts 3A, 3B. Although not shown, an interface layer may additionally be provided.

[0144] The helmet of FIG. 6 is configured such that the outer part 3A of the energy absorbing layer 3 is able to slide relative to the inner part 3B of the energy absorbing layer 3 in response to an impact. A sliding interface may be provided between the outer part 3A of the energy absorbing layer 3 and the inner part 3B of the energy absorbing layer 3.

[0145] Although not shown, an additional layer may be provided on a surface of one or both of the inner and outer parts 3A, 3B of the energy absorbing layer 3 facing the sliding interface. The additional layer may be moulded to the inner or outer parts 3A, 3B of the energy absorbing layer 3 or otherwise attached thereto. The additional layer may be formed from a relatively hard material, e.g. relative to the energy absorbing layer 3. The additional layer may be configured to provide friction reducing means to reduce the friction at the sliding interface. This may be achieved by forming the additional layer from a low friction material, such as PC, PTFE, ABS, PVC, Nylon, PFA, EEP, PE and UHMWPE. Alternatively, or additionally, this may be achieved by applying a low friction coating to the additional layer and/or applying a lubricant to the additional layer.

[0146] Alternatively or additionally, friction reducing means, to reduce the friction at the sliding interface, may be provided by forming one or both of the inner and outer parts 3A, 3B of the energy absorbing layer 3 from a low friction material, providing an additional low friction layer on a surface of the inner and outer parts 3A, 3B of the energy absorbing layer 3 facing the sliding interface, by applying a low friction coating to the inner and outer parts 3A, 3B of the energy absorbing layer 3 and/or applying a lubricant to the inner and outer parts 3A, 3B of the energy absorbing layer 3.

[0147] The helmet 1 shown in FIG. 6 also comprises connectors 5 attached to the outer layer 2. The connectors 5 are also attached to the energy absorbing layer 3 (or additional layer) to allow relative sliding between the energy absorbing layer 3 and the sections of the interface layer 4. Alternatively or additionally, one or more of the connectors

5 may be connected to another part of the remainder of the helmet 1, such as an interface layer. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0148] It should be understood that such an arrangement of inner and outer parts 3A 3B of the energy absorbing layer 3 may be added to any helmet described herein.

[0149] FIG. 11 shows a helmet substantially the same as the helmet shown in FIG. 6. However, in the helmet of FIG. 11, the inner part 3B of the energy absorbing layer 3 is formed in multiple parts 3C and 3D adjacent each other in the circumferential direction of the helmet. These parts 3C and 3D are configured to move relative to each other, as well as to the outer part 3A of the energy absorbing layer. The parts 3C and 3D may be connected to each other by one or more connectors that allow relative movement.

[0150] FIG. 7 shows a helmet comprising an outer layer 2 and an energy absorbing layer 3. As shown in FIG. 7, one or more outer plates 7 are mounted to the outer layer 2 of the helmet 1. The outer plates 7 may be formed from a relatively strong and/or rigid material, for example from the same types of materials as from which the outer layer 2 may be formed. Although not shown, an interface layer may additionally be provided.

[0151] The helmet of FIG. 7 is configured such that the outer plates 8 are able to slide relative to the outer layer 2 in response to an impact. A sliding interface may be provided between the outer plates 8 and the outer layer 2.

[0152] Friction reducing means, to reduce the friction at the sliding interface, may be provided by forming the outer layer 2 and/or the outer plates 8 from a low friction material, providing an additional low friction layer on a surface of the outer layer 2 and/or the outer plates 8 facing the sliding interface, by applying a low friction coating to the outer layer 2 and/or the outer plates 8, and/or applying a lubricant to the outer layer 2 and/or the outer plates 8.

[0153] The helmet 1 shown in FIG. 7 also comprises connectors 5 attached to the outer plates 7. The connectors 5 are also attached to the outer layer 2 to allow relative sliding between the plates 7 and the outer layer 2. Alternatively or additionally, one or more of the connectors 5 may be connected to another part of the remainder of the helmet 1, such as the energy absorbing layer 3. The connectors 5 may also be connected to two or more parts of the remainder of the helmet 1.

[0154] In such an arrangement, in the event of an impact on the helmet 1, it can be expected that the impact would be incident on one or a limited number of the outer plates 17. Therefore, by configuring the helmet such that the one or more outer plates 7 can move relative to the outer layer 2 and any outer plates 7 that have not been subject to an impact, the surface receiving the impact, namely one or a limited number of outer plates 7, can move relative to the remainder of the helmet 1. In the case of an impact, this may reduce the rotational acceleration of the head of a wearer.

[0155] It should be understood that such an arrangement of outer plates 7 may be added to any helmet described herein, namely an arrangement having a sliding interface between at least two of the layers of the helmet 1.

[0156] Some helmets, such as those shown in FIGS. 1 to 6, are configured to cover a top portion of the head and the above described helmet structures are appropriately located in the helmet to cover a top portion of the head. For example, a helmet may be provided to substantially cover the fore-

head, top of the head, back of the head and/or temples of the wearer. The helmet may substantially cover the cranium of the wearer.

[0157] Some helmets may be configured to cover other parts of the head, alternatively or additionally to a top portion. For example, helmets such as the helmet shown in FIG. 8 may cover the cheeks and/or chin of the wearer. Such helmets may be configured to substantially cover the jaw of the wearer. Helmets of the type shown in FIG. 8, are often referred to as full-face helmets. As shown in FIG. 8, cheek pads 30 may be provided on either side of the helmet 1 (i.e. left and right sides). The cheek pads 30 may be arranged within an outer shell 2 of the helmet 1 to protect the side of the face of the wearer from an impact.

[0158] The cheek pads 30 may have the same layered structure as the example helmets described above. For example, the cheek pads 30 may comprise one or more energy absorbing layers as described above, and/or an interface layer as described above, and/or layers that move relative to each other as described above, optionally, layers may be connected by connectors as described above. Alternatively or additionally, the cheek pads 30 themselves may be configured to move relative to the outer shell 2 and, optionally be connected to the outer shell by connectors as described above.

[0159] Although, the above examples relate to helmets, as stated above, the disclosure may also relate to alternative protective apparel, such as body armour, as shown in FIGS. 9 and 10. Body armour 100 may provide protection for other parts of the body, such as the shins, knees, thighs, forearms, elbows, upper arms, shoulders, chest and back. Individual items of body armour may be provided to protect individual body parts (as shown in FIG. 9), or alternatively may be combined in apparel comprising multiple armoured regions 101 to protect more than one body part (as shown in FIG. 10). Such body armour 100 may be worn for the same activities as helmets, discussed above, including for combat, sports, and motorcycling.

[0160] The body armour 100 may have the same layered structure as the example helmets described above. For example, the body armour 100 may comprise an outer shell 2 as described above, one or more energy absorbing layers 3 as described above, and/or an interface layer as described above, and/or layers that move relative to each other as described above, and/or layers may be connected by connectors 5 as described above.

[0161] Connectors that may be used within a helmet are described below. It should be appreciated that these connectors may be used in a variety of contexts and are not to be limited to use within helmets. For example, they may be used in other apparatuses that provide impact protection, such as body armour or padding for sports equipment.

[0162] It should be appreciated that the connectors may be used for connecting any two parts of an apparatus together. In the context of helmets, it should be appreciated in particular that the connectors may be used for connecting any two parts of helmets such as those discussed above that are configured to move relative to each other.

[0163] Where a connector is described as having a first part connected to a first part of an apparatus and a second part connected to a second part of an apparatus, it should be appreciated that, with suitable modifications, this may be reversed. It should also be appreciated that where an appa-

ratus has first and second parts connected by plural connectors, the plural connectors need not have the same configuration as each other.

[0164] FIGS. 12 and 13 show a first example connector 20 according to the disclosure. The connector 20 is for connecting first and second parts of a helmet, the first and second parts being configured to move relative to each other at least in a first plane. As shown, the connector comprises a first attachment part 21 for connecting to the first part of the helmet and a second attachment part 22 for connecting to the second part of the helmet. A resilient part 23 is provided extending between the first and second attachment parts 21, 22. The resilient part 23 is configured to deform to allow relative movement between the first and second attachment parts 21, 22.

[0165] As shown in FIGS. 12 and 13, the first attachment part 21 and the second attachment part 22 are displaced relative to each other in a first direction. As shown, the resilient part 23 is narrower, in a second direction perpendicular to the first direction, than the first attachment part 21 and the second attachment part 22.

[0166] As shown, the resilient part 23 may be narrower, in a third direction perpendicular to the first and second directions, than the first attachment part 21 and the second attachment part 22. Alternatively, the first attachment part 21, the second attachment part 22, and the resilient portion 23 may be substantially the same thickness in the third direction.

[0167] As shown in FIGS. 12 and 13, the first attachment part 21 and the second attachment part 22 may be substantially circular in shape, as viewed from the third direction. The first attachment part 21 and the second attachment part 22 may be substantially cylindrical. In other examples not shown, one or both of the first attachment part 21 and the second attachment part 22 may have a different shape.

[0168] As shown in FIG. 13, the first attachment part 21 and the second attachment part 22 may be substantially hollow. As shown, the first attachment part 21 and the second attachment part 22 may each comprise an outer wall 211, 221 and a base 212, 222, defining a hollow cavity. In other examples not shown, one or both of the first attachment part 21 and the second attachment part 22 may be solid.

[0169] As shown in FIGS. 12 and 13, the first attachment part 21 may be wider than the second attachment part 22, in the second direction. In other examples not shown, the reverse may be the case. Alternatively, the first attachment part 21 and the second attachment part 22 may be the substantially same width.

[0170] As shown in FIGS. 12 and 13, the resilient part 23, may be substantially elongate. As shown in FIGS. 12 and 13, the resilient part 23 may extend substantially linearly. In other examples, the resilient part 23 may extend along a substantially curved path. The resilient part 23 may be formed from a resilient material. The shape and material of the resilient part enable deformation that allows relative movement between the first attachment part 21 and the second attachment part 22.

[0171] The resilient part 23 may be formed from a resilient material. The resilient material may be a TPU (Thermoplastic Polyurethane) material, TPE (Thermoplastic elastomer) material, or a silicone elastomer material. As shown in FIGS. 12 and 13, the first attachment part 21 and the second attachment part 22 may also be formed from a resilient

material, e.g. the same resilient material as the resilient part 23. These three parts may be integrally formed, as shown in FIGS. 12 and 13.

[0172] FIG. 14 shows part of an example helmet 1 comprising the example connector 20. As shown, the helmet 1 comprises a first part 3C and a second part 3D configured to move relative to each other at least in a first plane, as indicated by arrows. As shown the first and second parts 3C, 3D may be arranged adjacent each other in the first plane. The first and second parts 3C, 3D may form a first layer 3B of the helmet 1. As shown, the first and second parts 3C, 3D comprise respective recesses 31.

[0173] As shown in FIG. 14 the first attachment part 21 and the second attachment part 22 may be configured to press fit into a corresponding recess 31 in the respective first or second part of the helmet 3C, 3D. The recesses 31 may be in communication with respective channels 32 in the first or second part of the helmet 3C, 3D in which the resilient part 23 is arranged.

[0174] As shown in FIG. 14, the connector may be configured to connect the first and second parts of the helmet 3C, 3D such that, in use, the first direction (the direction of displacement between the first and second attachment parts 21, 22) is substantially parallel to the first plane (in which the parts 3C, 3D of the helmet move relative to each other).

[0175] As shown in FIG. 14, the first attachment part 21 and the second attachment part 22 may be configured to be inserted into the corresponding recess 31 in the respective first or second part 3C, 3D of the helmet in a direction substantially perpendicular to the first plane, i.e. parallel to the third direction.

[0176] As shown in FIGS. 12 and 13 the first attachment part 21 and the second attachment part 22 may comprise a protrusion 24 configured to engage with the corresponding recess 31. As shown, this may extend radially outwards from the first attachment part 21. As shown in FIGS. 12 and 13, the protrusion 24 may comprise a sloped portion 25 and a shelf 26. The sloped portion 25 may be configured to ease insertion of the attachment part 21, 22 into the recess 31 and the shelf 26 may be configured to resist removal of the attachment part 21, 22 from the recess 31.

[0177] As shown in FIGS. 12 and 13, a single protrusion 24 may be provided. This may be provided continuously around an outer edge of the attachment parts 21, 22.

[0178] As shown in FIGS. 12 and 13, the first attachment part 21 may comprise an anchor point, in this example in the form of a through hole 213 in base 212, for connecting the connector to a third part of the helmet or a further connector. As shown in FIG. 15, the through hole 213 may receive a snap-pin 28. The snap-pin 28 may attach to a corresponding snap-basket 29 in a third part of the helmet 3A.

[0179] As shown in FIG. 15, the third part of the helmet 3A may be arranged adjacent the first and second parts 3C, 3D in a direction perpendicular to the first plane in which the first and second parts 3C, 3D move. The third part of the helmet 3A may form a second layer of the helmet 1.

[0180] As shown in FIG. 15, the connector 20 may be arranged on a side of the first and second parts 3C, 3D that is closest to the third part 3A. In other examples not shown, the connector 20 may be arranged on a side of the first and second parts 3C, 3D that is furthest from the third part 3A. The connectors of such examples may or may not include an anchor point as described above.

[0181] The third part of the helmet 3A may be configured to move relative to the first and second parts 3C, 3D in a second plane substantially parallel to the first plane. As shown in FIG. 15, an intermediate layer 4 of low friction material (as described in detail above) may be provided between the first and second layers 3B, 3A. The intermediate layer 4 may be provided on a surface of the first and second parts 3C, 3D. As shown, the intermediate layer 4 may partially, completely cover the recesses 31 in the first and second parts 3C, 3D.

[0182] As shown in FIG. 15, the first, second, and/or third parts of the helmet 3C, 3D, 3A may be energy absorbing parts, forming energy absorbing layers of the helmet (as described in detail above). In an example not shown, the third part may instead be an outer shell (as described in detail above) or an interface layer (as described in detail above).

[0183] FIG. 16 shows a second example connector 20 in which the first attachment part 21 differs from that of the first example connector shown in FIGS. 12 and 13. Specifically, the first attachment part 21 is connected to a third attachment part 213 for connecting to the third part of the helmet 3A.

[0184] As shown in FIG. 16, the second example connector 20 comprises a second resilient part 214 extending between the first attachment part 21 and the third attachment part 27 and configured to deform to allow relative movement between the third and first attachment parts, 27, 21. The third part of the helmet 3A may be configured to move relative to the first and second parts 3C, 3D in a second plane substantially parallel to the first plane.

[0185] As shown in FIG. 16, the first attachment part 21 may comprise a peripheral portion 215, surrounding the second resilient part 214 about a first axis. The second resilient part 214 may in turn surround the third attachment part 213 about a second axis (which may or may not be the same as the first axis).

[0186] As shown, the peripheral portion 215 may be annular in shape. As shown the second resilient part 214 may span substantially the entirety of a central portion of the first attachment part 21. Alternatively, the second resilient part 214 may be formed from one or more arms with space between, e.g. forming an X-shape in the profile view of FIG. 16.

[0187] The second resilient part 214 may be formed from a resilient material (as described above in relation to the first resilient part 23). As shown in FIG. 16, the peripheral portion may be formed from a relatively hard material compare to the resilient portion 214. These parts may be co-moulded together. As shown in FIG. 16, the first resilient part 23 may extend from the peripheral part 215. These parts may be co-moulded together.

[0188] As shown in FIG. 16, protrusions 24 may be provided to the peripheral portion 215. These may be formed from a resilient material. They may also be co-moulded with the peripheral portion 215.

[0189] FIG. 17 shows part of an example helmet 1 comprising the second example connector 20. The helmet is substantially the same as that shown in FIG. 15 and described above. As shown, the second resilient part 214 is configured to protrude from the peripheral portion 215 in a direction along the first axis, when the connector 20 is connected to the helmet 1.

[0190] As shown, the third attachment part 213 may be a through hole configured to allow a fastener, such as snap pin

28 to pass through, and connect to the third part 3A of the helmet, e.g. via snap basket 29.

[0191] The connector shown in FIGS. 16 and 17 connects all three helmet parts 3A, 3C, 3D and allows independent relative movement between each, at least in the first and second planes.

[0192] FIGS. 18 and 19 show a variation of the connector 20 shown in FIGS. 12 and 13. The variations may also be applied to the connector shown in FIG. 16.

[0193] As shown in FIG. 18, the resilient part 23 may vary in thickness in the third direction. For example, as shown, the resilient part 23 may comprise a relatively thin portion 231 and a relatively thick portion 232. The relatively thin portion 231 may comprise tapered portions where the thickness changes gradually. This may allow the connector to bend more easily at the relatively thin portion 231 in a plane parallel to the third direction.

[0194] As shown in FIG. 19, the resilient part 23 may vary in width in the second direction. For example, as shown, the resilient part 23 may comprise a relatively narrow portion 233 and a relatively wide portion 234. The relatively narrow portion 233 may comprise tapered portions where the thickness changes gradually. This may allow the connector to bend more easily at the relatively thin portion 231 in a plane parallel to the second direction.

[0195] As shown, the relatively thin portion 231 may be located in the relatively wide portion 234 of the resilient part 23.

[0196] As shown in FIG. 19, the protrusion 24 is divided into a plurality of protrusions 24. These are provided adjacent each other around an outer edge of the attachment parts 21, 22.

[0197] FIGS. 20 to 22 show examples of how the arrangements of the first and second attachment parts 21, 22 may be varied, e.g. for the above described connectors and variations thereof.

[0198] As shown in FIG. 20, the first attachment part 21 may be relatively thin in the first direction, e.g. so as to be arranged only at portion of an edge of the corresponding recess 31. This may allow room for an additional connector 5 within the central part of the recess 31, as shown. The additional connector 5 may connect the first part of the helmet 3C to the third part of the helmet 3A.

[0199] As shown in FIG. 21, both the first and second attachment parts may have this arrangements. The thickness of the first and/or second attachment parts 21, 22 in the first direction may be similar to the thickness of the resilient part 23 in the second direction, for example.

[0200] As shown in FIG. 22, the first and second attachment parts may be substantially the same size.

[0201] FIGS. 23 to 27 show examples of how the arrangements of the resilient part 23 may be varied, e.g. for the above described connectors and variations thereof.

[0202] As shown in FIG. 23, the resilient part 23 may extend on a curved path in the first direction, e.g. rather than on a straight path as in other examples. This path may be sinusoidal as viewed from the third direction, as shown.

[0203] As shown in, FIG. 24, the resilient part 23 may comprise relatively narrow portions 233 and relatively wide portions 234 in the second direction.

[0204] As shown in FIG. 25, the resilient part 23 may be thinner in the third direction than it is wide in the second direction. Conversely, as shown in FIG. 26, the resilient part 23 may be thicker in the third direction than it is wide in the

second direction. As shown in FIG. 27, the resilient part may comprise portions that are thinner in the third direction than wide in the second direction and portions that are thicker in the third direction than wide in the second direction.

[0205] FIG. 28 shows how the arrangements of the protrusions 24 may be varied. As shown multiple rows of protrusions 24 may be provided arranged adjacent each other in the third direction.

[0206] FIG. 29 shows an example in which multiple connectors 20 with corresponding recesses 31 and channels, are used to connect multiple parts 3X of a helmet.

[0207] FIGS. 30 to 32 show how the connectors described above may be further varied by comprising multiple second attachment parts 23 and multiple corresponding resilient parts 23. FIGS. 30 to 32 respectively show connectors with four, three, or two second attachment parts 23 and multiple corresponding resilient parts 23. In the examples, the multiple resilient parts 23 extend from a single first attachment part 21. The multiple second attachment parts 22 may be configured to connect to multiple respective parts of a helmet, for example. However, they may alternatively provide multiple connections to fewer helmet parts.

[0208] FIG. 33 shows an example in which a connector with multiple second attachment parts 23 and multiple corresponding resilient parts 23, with corresponding recesses 31 and channels, is used to connect multiple respective parts 3X of a helmet.

[0209] When connectors 20 such as those described above are used in a helmet of the type shown in FIG. 11, the connectors 20 allow relative movement between first and second parts 3C, 3D while securing said parts together. Some example connectors 20 additionally allow relative movement between the first and second parts 3C, 3D and a third part 3A, while securing said parts together. However, it should be understood that the connectors described are not limited to use within this type of helmet.

[0210] Helmets as described above may be used in various activities. These activities include combat and industrial purposes, such as protective helmets for soldiers and hardhats or helmets used by builders, mine-workers, or operators of industrial machinery for example. Helmets, are also common in sporting activities. For example, protective helmets may be used in ice hockey, cycling, motorcycling, motor-car racing, skiing, snowboarding, skating, skateboarding, equestrian activities, American football, baseball, rugby, soccer, cricket, lacrosse, climbing, golf, airsoft, roller derby and paintballing.

[0211] Examples of injuries that may be prevented or mitigated by the helmets described above include Mild Traumatic Brain Injuries (MTBI) such as concussion, and Severe Traumatic Brain Injuries (STBI) such as subdural haematomas (SDH), bleeding as a consequence of blood vessels rupturing, and diffuse axonal injuries (DAI), which can be summarized as nerve fibres being over stretched as a consequence of high shear deformations in the brain tissue.

[0212] Depending on the characteristics of the rotational component of an impact, such as the duration, amplitude and rate of increase, either concussion, SDH, DAI or a combination of these injuries can be suffered. Generally speaking, SDH occur in the case of accelerations of short duration and great amplitude, while DAI occur in the case of longer and more widespread acceleration loads.

[0213] Variations of the above described examples are possible in light of the above teachings. It is to be understood

that the invention may be practiced otherwise and specifically described herein without departing from the spirit and scope of the invention.

1. A connector for connecting first and second parts of an apparatus, the first and second parts being arranged adjacent each other in a first plane and configured to move relative to each other at least in the first plane, the connector comprising:

- a first attachment part for connecting to the first part of the apparatus;
- a second attachment part for connecting to the second part of the apparatus;
- a resilient part extending between the first and second attachment parts and configured to deform to allow relative movement between the first and second attachment parts in the first plane, wherein:
 - the first attachment part and the second attachment part are displaced relative to each other in a first direction, the first direction being parallel to the first plane when in use, and
 - the resilient part is narrower, in a second direction perpendicular to the first direction, than the first attachment part and the second attachment part.

2. The connector of claim 1, wherein:

at least one of the first attachment part and the second attachment part is configured to press fit into a corresponding recess in the respective first or second part of the apparatus.

3. The connector of claim 2, wherein:

the first attachment part and the second attachment part are displaced relative to each other in a first direction, and the connector is configured to connect the first and second parts of the apparatus such that, in use, the first direction is substantially parallel to the first plane, and the at least one of the first attachment part and the second attachment part is configured to be inserted into the corresponding recess in the respective first or second part of the apparatus in a direction substantially perpendicular to the first plane.

4. The connector of claim 2, wherein the at least one of the first attachment part and the second attachment part comprises one or more protrusions configured to engage with the corresponding recess, optionally wherein the one or more protrusions comprises a sloped portion and a shelf, the sloped portion being configured to ease insertion of the attachment part into the recess and the shelf configured to resist removal of the attachment part from the recess.

5. (canceled)

6. The connector of claim 1, wherein the first attachment part and/or the second attachment part are formed from a resilient material, optionally wherein the resilient part is formed from the same resilient material.

7. (canceled)

8. The connector of claim 1, wherein at least one of the first and second attachment parts comprises an anchor point configured to connect the connector to a third part of the apparatus or a further connector, optionally wherein the anchor point comprise a hole configured to receive a snap-pin.

9. (canceled)

10. The connector of claim 1 wherein the first and/or second attachment part is substantially circular.

11. The connector of claim 1, wherein the resilient part is elongate.

12. The connector of claim 1, comprising multiple second attachment parts, and multiple respective resilient parts, for connecting to one or more second parts of the apparatus.

13. The connector of claim 1, further comprising a third attachment part for connecting to a third part of the apparatus, the third part of the apparatus being configured to move relative to the first and second parts in a second plane substantially parallel to the first plane; and

- a second resilient part extending between the first attachment part and the third attachment part and configured to deform to allow relative movement between the third and first attachment parts.

14. A connector for connecting first, second and third parts of an apparatus, the first and second parts being arranged adjacent each other in a first plane and configured to move relative to each other at least in the first plane, and the third part being configured to move relative to the first and second parts in a second plane substantially parallel to the first plane, the connector comprising:

- a first attachment part for connecting to the first part of the apparatus;
- a second attachment part for connecting to the second part of the apparatus;
- a third attachment part for connecting to the third part of the apparatus;
- a first resilient part extending between the first and second attachment parts and configured to deform to allow relative movement between the first and second attachment parts in the first plane and;
- a second resilient part extending between the first attachment part and the third attachment part and configured to deform to allow relative movement between the third and first attachment parts in the first and second planes.

15. The connector of claim 14, wherein the first and third attachment parts are detachable from each other, such that the connector is formed in two parts.

16. The connector of claim 14, wherein the first attachment part comprises a peripheral portion, surrounding the second resilient part about a first axis, and the second resilient part surrounds the third attachment part about a second axis, and

- the resilient portion is configured to protrude from the peripheral portion in a direction along the first axis, when the connector is connected to the apparatus.

17. An apparatus comprising:

- a first part and a second part, the first and second parts being arranged adjacent each other in a first plane and configured to move relative to each other at least in the first plane;
- a connector connecting first and second parts of an apparatus, the connector comprising:
 - a first attachment part for connecting to the first part of the apparatus;
 - a second attachment part for connecting to the second part of the apparatus;
 - a resilient part extending between the first and second attachment parts and configured to deform to allow relative movement between the first and second attachment parts in the first plane, wherein:
 - the first attachment part and the second attachment part are displaced relative to each other in a first direction, the first direction being parallel to the first plane when in use, and

the resilient part is narrower, in a second direction perpendicular to the first direction, than the first attachment part and the second attachment part, wherein the first attachment part of the connector is connected to the first part of the apparatus and the second attachment part of the connector is connected to the second part of the apparatus.

18. The apparatus of claim **17**, wherein the first part comprises a first recess and the first attachment part is arranged in the first recess, and/or the second part comprises a second recess and the second attachment part of the connector is arranged in the second recess,

optionally wherein the first part and/or the second part comprise a channel in communication with a respective recess, wherein the resilient part is arranged in the channel,

optionally wherein the first and second recesses are substantially the same shape as the respective attachment parts, and/or wherein at least one of the first and second attachment parts is press fit into the respective recess.

19. (canceled)

20. (canceled)

21. (canceled)

22. The apparatus of claim **18**, wherein the connector further comprises a third attachment part for connecting to a third part of the apparatus, the third part of the apparatus being configured to move relative to the first and second parts in a second plane substantially parallel to the first plane, and

a second resilient part extending between the first attachment part and the third attachment part and configured to deform to allow relative movement between the third and first attachment parts, and

the third attachment part is also arranged in the first recess.

23. The apparatus of claim **17**, wherein the apparatus has a layered structure and the first and second parts together form a first layer.

24. (canceled)

25. The apparatus of claim **17**, wherein the apparatus further comprises a third part configured to move relative to the first and second parts in a second plane substantially parallel to the first plane, optionally wherein:

the connector further comprises a third attachment part for connecting to a third part of the apparatus, the third part of the apparatus being configured to move relative to the first and second parts in a second plane substantially parallel to the first plane, and

a second resilient part extending between the first attachment part and the third attachment part and configured to deform to allow relative movement between the third and first attachment parts, and

the third attachment part is connected to the third part of the apparatus;

optionally wherein:

the apparatus has a layered structure and the first and second parts together form a first layer and the third part forms a second layer adjacent the first layer.

26. (canceled)

27. (canceled)

28. The apparatus of claim **17**, wherein the first and second parts are energy absorbing parts, optionally forming a first energy absorbing layer, optionally wherein the connector further comprises a third attachment part for connecting to a third part of the apparatus, the third part of the apparatus being configured to move relative to the first and second parts in a second plane substantially parallel to the first plane, and

a second resilient part extending between the first attachment part and the third attachment part and configured to deform to allow relative movement between the third and first attachment parts, and

the third part is an energy absorbing part, optionally forming a second energy absorbing layer.

29. (canceled)

30. The apparatus of claim **17**, wherein the apparatus is a helmet.

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